Automated Detection of Grooming Strategies in Online Chat Conversations:

A Fine-Tuned Large Language Model Approach

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Abstract

Introduction Online grooming is a critical cyber threat in which adults with malicious intent systematically manipulate minors through digital platforms to facilitate sexual abuse or exploitation. Current automated detection systems primarily rely on binary classification, limiting their ability to identify the specific manipulation strategies required by child protection professionals for targeted intervention. Emerging technologies such as Large Language Models offer promising, yet largely unexplored, opportunities to address this gap.

Research Question "How effectively can finetuned large language models identify specific grooming strategies in online chat conversations?"

Method This study employs a mixed-methods approach, combining the Design Science Research strategy and a quasi-experimental evaluation. Using the PAN2012 Sexual Predator Identification corpus, 200 grooming conversations were manually annotated with a six-step framework that identifies five grooming strategies, including Sexual Desensitization, Sexual Normalization, Isolation and Secrecy, Sexual Solicitation and Exploitation, and Control and Coercion, based on a conversational arc principle that captures what predators work toward over the course of an interaction. Two annotators independently labeled primary and secondary strategies to build a high quality training set. Cross-platform generalization was examined using 40 additional conversations from the LiveMe Grooming Dataset, selected via a keyword-based scoring script. Model performance was evaluated under two conditions: Condition A, a strategy-informed prompting baseline using the unmodified Gemma 3-4B instruction-tuned model, and Condition B, the same model fine-tuned with Low Rank Adaptation on 120 annotated conversations in full bfloat16 precision. This design isolates fine tuning as the sole independent variable, enabling a controlled assessment of its impact on grooming strategy classification.

Results Inter-annotator agreement reached Cohen's kappa of 0.874 for primary strategy annotation on PAN2012 and 0.787 on LiveMe, both exceeding the 0.7 reliability threshold, confirming the annotation framework produces consistent and credible ground truth. Fine-tuning improved the Gemma 3-4B model accuracy from 40.0% to 52.5% and macro F1 from 0.214 to 0.256. The most significant improvement was in Sexual Normalization, which rose from F1 of 0.000 under the baseline to 0.400 after fine-tuning. A supplementary Gemma 3-12B evaluation achieved stronger baseline performance with a macro F1 of 0.407, highlighting the benefits of larger model capacity for prompt-based detection. Cross-platform evaluation on the LiveMe Grooming Dataset without further training yielded macro F1 scores of 0.337 for the 4B model and 0.435 for the 12B model, demonstrating meaningful generalizations of learned strategy representations across platforms with substantially different communicative contexts.

Discussion The findings confirm that fine-tuned large language models can identify specific grooming strategies effectively, answering the research question affirmatively. Cross-platform transfer to LiveMe demonstrates that learned strategy representations are not dataset-specific. However, modest absolute performance levels, small per-class sample sizes for rare strategies, and the ecological limitations of PAN2012 decoy conversations constrain the generalizability of these results. Future research should focus on expanding the annotated datasets, fine-tuning on large models with full precision and multi-modal detection to improve the level of grooming classification toward practical deployment.

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Finally, we recognize that this research addresses sensitive issues concerning child safety, and we are humbled to have been able to contribute to protecting children in the digital environment.

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BoW	Bag of Words
CEOP	Child Exploitation and Online Protection Centre
CNN	Convolutional Neural Network
DSR	Design Science Research
eSPD	Early Sexual Predator Detection
FRFS	Fuzzy-Rough Feature Selection
GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
IRC	Internet Relay Chat
k-NN	k-Nearest Neighbours
LLM	Large Language Model
LLMs	Large Language Models
LoRA	Low-Rank Adaptation
ML	Machine Learning
NCMEC	National Center for Missing and Exploited Children
NLP	Natural Language Processing
OCR	Optical Character Recognition

OSINT	Open Source Intelligence
OSNs	Online Social Networks
PCA	Principal Component Analysis
PCI	Predatory Conversation Identification
PEFT	Parameter-Efficient Fine-Tuning
QLoRA	Quantised Low-Rank Adaptation
RoBERTa	Robustly Optimised BERT Pre-training Approach
SimCSE	Simple Contrastive Sentence Embedding
SSH	Secure Shell
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
VPD	Victim-from-Predator Distinction
VRAM	Video Random Access Memory
XML	Extensible Markup Language

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1

Introduction

The evolution of digital communication platforms has revolutionized how children and teens communicate and relate through online platforms, going beyond the sphere of family and school. These platforms offer immense opportunities for communication, learning, and socialization, but have also opened channels for the exploitation of minors by predatory adults. Online grooming, which can be defined as the process through which an adult establishes an emotional connection with a minor via online platforms, with the ultimate objective of sexual exploitation [1, 2]. This has emerged as one of the most pressing child safety concerns of the digital age. Recent studies have revealed that approximately one in four adolescents have encountered online harassment or inappropriate content [3]. Also, some studies have shown that nearly half of offenders who were convicted for contact sexual offenses had previously displayed grooming behaviors online [4]. The severity of this problem is further emphasized by reports from the National Center for Missing and Exploited Children (NCMEC), which documented over 45 million images and videos of child sexual abuse in a single year, with figures doubling annually [1].

Detecting online grooming via conversations presents substantial technical and methodological challenges. Grooming is a multi-stage process that can take place over a long period of time, often starting with seemingly benign conversations that progress towards sexual abuse [2, 5]. The linguistic patterns employed by predators are diverse, contextualized, and often camouflaged within seemingly innocent conversation, making the task of detecting it a complex Natural Language Processing (NLP) challenge

[1, 6]. Traditional detection approaches, including keyword-based filtering and rule-based systems, have proven inadequate for capturing the sophisticated and evolving strategies used by online predators [7, 8]. More recent machine learning methods, while achieving high accuracy in binary classification of predatory versus non-predatory conversations, have largely focused on identifying whether a conversation is grooming-related rather than understanding the specific manipulation strategies employed within those conversations [1, 9].

Recent advances in large language models (LLMs) and transformer-based architectures have demonstrated exceptional capabilities in understanding contextual language and capturing long-range dependencies within text [6, 9, 10]. These developments offer a potential avenue to move beyond binary grooming detection toward the more granular task of identifying specific grooming strategies within conversations. This thesis investigates the effectiveness of fine-tuning a large language model, specifically Gemma 3[11, 12], on manually annotated grooming conversations to classify distinct manipulation strategies, thereby contributing to both the theoretical understanding of online grooming behavior and the practical development of more sophisticated child protection tools.

1.1 Research Problem

The existing research on automated grooming detection has predominantly concentrated on binary classification tasks, distinguishing predatory conversations from non-predatory ones, or identifying predatory authors within chat datasets [1, 5, 13] . While these approaches have achieved impressive performance metrics, with state-of-the-art systems reporting F1-scores exceeding 0.97 on the widely used PAN2012 benchmark dataset [6, 9], they treat grooming as a uniform phenomenon rather than as a collection of distinct manipulative strategies. This represents a significant limitation because grooming, as theorized by O'Connell [2] and elaborated through subsequent research [1], is a structured process comprising multiple stages and strategies, each serving different psychological objectives in the predator's manipulation of the victim.

The absence of strategy-level detection capabilities means that current systems can flag a conversation as potentially predatory but cannot characterize how the predator is attempting to manipulate the victim. This distinction is critically important for child protection practitioners, law enforcement agencies, and platform moderators who need to understand the

nature and severity of detected threats to prioritize responses appropriately. For instance, a conversation dominated by sexual solicitation and exploitation tactics may require more immediate intervention than one still in early desensitization stages [1, 2].

Finally, researchers still have not focused on the cross-platform generalization challenge. Most detection models are developed and evaluated exclusively on the PAN2012 dataset [1, 5, 9, 13, 14], which was constructed from Perverted Justice sting operations and IRC chat logs. The extent to which models trained on this dataset can generalize to grooming conversations occurring on modern social media platforms with substantially different communicative conventions remains largely unexplored [1, 4]. Borj et al. identified this cross-platform generalization gap as a major open problem, and Lykousas and Patsakis demonstrated that grooming on live streaming platforms like LiveMe exhibits distinct characteristics that may not be captured by models trained exclusively on older chat data [1, 4].

This study addresses these gaps by developing a fine-tuned LLM capable of classifying five specific grooming strategies within online conversations and evaluating its generalization capability across datasets derived from different communication platforms.

1.2 Research Aim and Question

The primary aim of this research is to investigate whether fine-tuning a large language model on manually annotated grooming conversations can effectively identify specific grooming strategies, moving beyond the binary detection paradigm that dominates current research. The study seeks to determine whether the contextual understanding afforded by transformer-based language models, combined with human-annotated training data grounded in established grooming theory, enables accurate classification of distinct manipulation tactics within online conversations. Such detection capabilities could enable precise intervention strategies, enhance investigator efficiency, improve understanding of predator behaviors, and strengthen training programs for child protection professionals.

Thus, this study's research question is as follows: **"How effectively can finetuned large language models identify specific grooming strategies in online chat conversations?"**

1.3 Delimitation of the Study

This study is delimited in several aspects to maintain methodological thoroughness and research feasibility within the constraints of a master's thesis. First, the study will focus exclusively on text-based grooming conversations in English. Even though grooming can occur across multiple languages and media, including voice, video, and image-based interactions [10], due to the scarcity of data logs and limited time duration, this study will only focus on text-based chat conversations. Another reason for limiting the conversation to a single language was that the study team was mostly fluent in English. This study has a phase of strategy annotation by human coders, which requires linguistic and cultural expertise that would multiply exponentially across multiple languages. But this methodological framework could potentially be extended to other modalities in future works.

Second, the primary dataset used for model training and evaluation is the PAN2012 corpus [1, 5]. This dataset comprises conversations between adult decoys posing as children and actual predators, rather than authentic predator-victim interactions. While this introduces a degree of ecological limitation, as noted by Vogt et al. [5] and Lykousas and Patsakis [4]. However, the PAN2012 dataset is the most common and publicly available dataset for grooming detection studies, and using it allows comparison with existing research.

Third, the study focuses on five late-stage grooming strategies: sexual desensitization, sexual normalization, isolation and secrecy, sexual solicitation and exploitation, and control and coercion. These strategies were selected based on established grooming theory [2] and the characteristics observable in the available data. Earlier grooming stages, such as friendship forming, relationship building, and risk assessment, while theoretically important, are deliberately excluded because they overlap substantially with ordinary online conversation patterns and are less amenable to reliable manual annotation at the conversation level [1, 2].

Fourth, the research develops and evaluates detection models for post-hoc analysis rather than real-time prediction, distinguishing it from real-time detection systems such as SafECon [3]. All data consists of anonymized conversations from existing research datasets, with no involvement of actual minors or active predators.

Fifth, the manual annotation phase focuses on specific grooming strategies identified through established research frameworks. While comprehensive taxonomies encompass various forms of online harassment, this study targets the core manipulation strategies most relevant to child protection applications, building upon the theoretical foundations established in early cyberstalking and grooming research [2].

Sixth, the computational resources available for this study have constraints with both scale of model that can be fine-tuned and extent of hyperparameter exploration. Usually, a larger LLM means larger requirements of GPU memory and processing power. Full-precision fine-tuning was comfortably feasible for the smaller Gemma 3-4B variant, but using larger 12B model required quantization to 4-bit precision[11].

Finally, the LiveMe Grooming Dataset [13] originates from a multi-user live streaming platform where chat sessions may involve several participants simultaneously. However, this study extracts and analyses only two-person conversations from that dataset. The primary basis of this study is the PAN2012 corpus, which consists of dyadic conversations between two users [3, 7]. The entire research design, including the manual annotation framework, strategy labeling process, and model fine-tuning pipeline, is built around this two-person predator-versus-victim structure. Therefore, to maintain consistency across both datasets, the same two-person filtering is applied to the LiveMe data.

2

Extended Background

This chapter provides a comprehensive review of the theoretical foundations, methodological evolution, and current state of research in automated online grooming detection. The chapter begins with a thorough review of the development of online grooming detection, from its psychological and legal origins to more sophisticated computational approaches. Next, investigate the latest trends in applying large language models and transformer architectures to this research field.

2.1 Theoretical Foundations and Early Research of Online Grooming

2.1.1 Psychological and Legal Foundation

The study of automated online grooming is anchored with psychological research that aims to understand the behavioral patterns and motivations of predators who use digital communication platforms to target minors. One of the fundamental contributions to this topic was made by O'Connell, who created a typology of child exploitation and online grooming practices [2]. She used a systematic analysis to study the interactions between adults and children in chat rooms. Her study revealed that online grooming is a deliberate, structured process where predators use specific strategies to gain access to children, build trust, assess risk, and ultimately achieve sexual exploitation [2].

Borj et al. [1] provided an extensive review of the psychological perspectives underlying online grooming research. According to their survey predators have a variety of motivations and strategies, from contact-driven offenders who seek to meet victims in person to fantasy-driven offenders who only exploit victim online. This distinction, originally identified through statistical discourse analysis of predatory conversations, is crucial for detection systems to understand the use of separate linguistic patterns and methods [1]. Furthermore, this study highlighted the complexity of grooming detection, which could be influenced by variations in duration, type, and intensity, making it challenging to predict the grooming process [1].

The legal landscape of online grooming has evolved alongside the psychological insights. Isaza et al. [14] documented the key legislative efforts across multiple jurisdictions, such as the United States Commission on Child Online Protection (Copa 2000) aimed at safeguarding minors from harmful online content, while the United Kingdom created the Child Exploitation and Online Protection Centre (CEOP) in 2006. Kim et al. [10] describe how South Korea's investigative tools in 2021 were used to address child sexual crimes, and also include the practical challenges that these measures have faced regarding their effectiveness. Vogt et al. [5] reveal that in England and Wales, over 5,000 offences of sexual communication with minors were logged with the police in a single year, averaging 14 offences per day, while Germany had experienced a 50% increase in recorded cases of child sexual abuse through internet communication. These alarming statistics, often regarded as underestimations due to under-reporting, highlight the pressing necessity for an automated detection solution in this domain [5].

2.1.2 Evolution of Digital Communication Platforms

During the past two decades, a large increment in the number of platforms with the possibility of anonymized communication. O'Connell's [2] original work focused on public chat rooms, which were the predominant medium for online interaction between strangers in the early 2000s. The shift from public to private communication has now created a critical escalation point that has reduced the likelihood of third-party detection.

The research by Lykousas and Patsakis highlighted the emergence of social live streaming services as a new frontier for predatory behavior [4]. With the modern social networks, who provide users with capabilities that

extend far beyond text-based chat, including live video broadcasting, virtual gift mechanisms, and real-time audience interaction ,introduced new vectors for predators that were not contemplated in earlier research. Importantly, their work demonstrated that the moderation systems deployed by the LiveMe platform were ineffective in detecting and suspending accounts engaged in predatory behavior, highlighting the gap between platform safety measures and the sophistication of predatory tactics [4].

The evolution of communication platforms has also expanded the demographic of potential victims and changed the dynamics of predator-victim interaction. Hamm [9] noted that surveys from 2023 showed 96% of American teenagers aged 13 to 17 used social media platforms daily, and that many platforms do not effectively verify age during registration. Sallahuddin et al. [3] emphasized that the pervasive use of online communication tools has made the problem increasingly urgent, with vulnerable children facing risks of grooming, harassment, and exposure to explicit content across an ever-expanding array of digital platforms. Kim et al. [10] further documented that in South Korea, approximately 10% of adolescents reported having met strangers encountered online in person, illustrating the real-world consequences.

2.1.3 Grooming Process Stages and Strategy Identification

O'Connell [2] proposed a model comprising six stages: friendship forming, relationship forming, risk assessment, exclusivity, sexual stage, and conclusion. Distinct communicative behaviors and psychological objectives characterize each stage. During the friendship-forming stage, the predator makes initial contact and gathers basic information about the minor. Then it is followed by the relationship-forming stage, which enhances the connection through discussions of interests and emotional bonding. In the risk assessment stage the predator evaluates detection risks by guardians. The exclusivity stage establishes mutual secrecy and connection, while the sexual stage introduces and escalates sexual content. The conclusion stage may lead to a physical meeting or ongoing online exploitation [2].

Borj et al. [1] provided a comprehensive synthesis of how these stages have been operationalization in computational research. Their survey revealed that researchers have conceptualized the grooming process as consisting of anywhere from three to six stages, with different studies emphasizing different aspects of the process. The three-stage models typically distinguish between gaining access, developing a deceptive relationship, and the sexual affair, while six-stage models incorporate the full taxonomy

proposed by O'Connell [1, 2]. Importantly, the study highlighted that the grooming stages are not necessarily sequential; predators may return to earlier stages to maintain the victim's trust, skip stages entirely, or spend varying amounts of time at each stage depending on their personality, motivations, and the victim's responses.

Hamm [9] further connected these grooming strategies to social exchange theory, arguing that predators manipulate the victim's cost-benefit calculus by providing emotional rewards such as attention, validation, and a sense of specialness, while gradually escalating the costs of the relationship through shame, guilt, and isolation from support networks.

Despite the well-established theoretical understanding of grooming stages and strategies, the computational literature has mainly applied these frameworks to binary or stage-level classification rather than to the identification of specific manipulation strategies within the later stages of grooming. This study builds upon the existing stage-based frameworks by developing a five-strategy taxonomy focused on the late-stage manipulation tactics that are most relevant for child protection intervention.

2.2 Evolution of Automated Detection Methodologies

2.2.1 Early Rule-based Approaches

The initiation approach for an automated grooming detection relied on keyword-based searching and rule-based systems. Anderson et al. described how investigators traditionally used keyword searches within digital forensic tools to identify grooming conversations, searching for specific text strings or lists of words associated with predatory behavior [7]. These tools suffered from several fundamental limitations, such as keyword searches frequently returning large volumes of irrelevant information, wasting the investigator's time. They could not group or order results in ways that supported efficient review. Furthermore, the selection of keywords was inherently subjective, introducing noise into the detection process [7]. Borj et al. [1] documented how early rule-based approaches also included phrase-matching techniques and dictionary-based methods for classifying messages into grooming stages. While these approaches provided a useful starting point for stage-level analysis, they were fundamentally limited by their reliance on predetermined lexicons that could not adapt to the evolving and diverse language used by predators across different platforms and contexts.

2.2.2 Traditional Machine Learning Methods

The transition from rule-based approaches to machine learning methods marked a major advancement in grooming detection capabilities. Ngejane et al. surveyed machine learning technologies applied to online sexual grooming detection, documenting the use of Support Vector Machines (SVM), k-Nearest Neighbours (k-NN), Naïve Bayes, logistic regression, Random Forest, and Decision Tree classifiers [8]. It has also acknowledged the early work by conducted by Pendar, showing k-NN models based on trigrams could achieve 94% accuracy in detecting predatory content, outperforming SVM models that achieved 90% accuracy [8]. However, performance degraded substantially when unigrams and bigrams were used, suggesting that the discriminative patterns in grooming conversations exist at higher-order textual structures.

Anderson et al. employed Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction, combined with fuzzy-rough feature selection (FRFS) for dimensionality reduction, to build discriminative document representations [7]. Their framework evaluated multiple classifiers and normalization techniques, demonstrating that feature extraction techniques played a crucial role in the performance by removing redundant and noisy features from the extracted representations [7].

Fauzi and Bours [13] proposed a two-stage classification approach that became influential in subsequent research. The first stage focused on Predatory Conversation Identification (PCI), distinguishing predatory from non-predatory conversations, while the second stage addressed Victim-from-Predator Distinction (VPD), identifying which participant in a predatory conversation was the predator. Using an ensemble method combining multiple classifiers with BoW features and TF-IDF weighting, they achieved an F0.5-score of 0.9348 on the PAN2012 dataset, a result that would have placed first in the original PAN2012 competition [13]. Their pre-filtering strategy, which retained only two-person conversations with at least six messages per user, reflected the understanding that predatory behavior requires sustained interaction between two participants.

Borj et al. [6] advanced the feature extraction paradigm by proposing a contrastive learning framework using RoBERTa encoders and supervised Simple Contrastive Learning of Sentence Embeddings(SimCSE) for sentence-level feature extraction. This approach addressed a key limitation of traditional BoW and TF-IDF methods. It eliminated the inability to handle unseen words, slang, and misspellings that are prevalent in online

chat conversations. When combined with SVM classification, this method achieved an F1-score of 0.96 and an accuracy of 0.99 for the detection of predatory conversations, benchmarking the state-of-the-art [6].

2.2.3 Deep Learning Advances

Deep learning techniques expanded the horizons in the field of grooming detection by introducing the capability to automate feature learning, reducing dependence on manually engineered features. Isaza et al. [14] developed a computational model combining Natural Language Processing with Convolutional Neural Networks (CNN) for classifying cybergrooming attacks. Their model processed messages from the PAN2012 corpus through a pre-processing pipeline that included tokenization, vocabulary analysis, and the construction of a trained word embedding matrix for semantic representation. While their model achieved high accuracy, it also produced a substantial number of false positives, reflecting the inherent challenge of detecting grooming messages that constitute a very small proportion of overall conversation content [14].

Vogt et al. [5] introduced the task of early Sexual Predator Detection (eSPD), re-framing the focus from forensic identification of completed grooming chats to proactive detection of grooming attempts during live conversations. Their methodology featured a two-tier architecture with the first tier utilizing fine-tuned BERT models to classify sliding windows of chat messages, and the second tier assesses the sequence of these classifications to determine when enough evidence is present to issue an alert. The researchers evaluated three variants of BERT, including MobileBERT, and demonstrated that BERT-base provided the optimal trade-off between early detection and classification accuracy, achieving a median warning latency of 18 messages and an F1-score of 0.89. However, it should be noted that these high performance scores are only achieved on binary classification tasks to identify predatory vs non-predatory. But our study addresses the identification of multiclass strategies. which is much more complicated process. Furthermore, most evaluations are conducted on the same PAN2012 dataset for training, and cross-dataset generalization still remains largely untested [1, 4].

Kim et al. [10] extended deep learning approaches to address language-specific challenges in grooming detection. They proposed a framework enhancing grooming detection in non-English languages by integrating Optical Character Recognition (OCR) with KcELECTRA, a deep learning NLP model tailored for colloquial Korean. They trained their model on a large

dataset of Korean social media conversations, achieving an accuracy of 0.953. This research emphasizes the necessity of adapting detection models to specific linguistic and cultural contexts.

2.3 Large Language Models (LLMs) and Transformer Architecture

2.3.1 Transformer Architecture and Attention Mechanisms

The transformer architecture, introduced by Vaswani et al. and subsequently adapted across numerous NLP applications, has become the foundation for modern language understanding systems. Borj et al. [6] explain that models like BERT use bidirectional training to learn contextual representations of text. Unlike earlier sequential models, BERT processes both left and right context simultaneously across all layers [6, 3]. The key innovation is the self-attention mechanism, which Hamm [9] describes as enabling transformers to capture dependencies over large text sequences by processing tokens in parallel rather than sequentially.

This parallel processing is especially valuable for grooming detection, where relationships between early and late messages can signal predatory intent [9, 5]. Several transformer variants have been applied to this task. Borj et al. [6] used RoBERTa within a contrastive learning framework to extract sentence-level embeddings. Vogt et al. [5] fine-tuned BERT-base, BERT-large, and MobileBERT as feature extractors for both conventional and early detection. Sallahuddin et al. [3] found that BERT can detect grooming patterns even in informal language with abbreviations and slang, which is critical for real-world deployment.

2.3.2 Pre-training and Transfer Learning

Hamm [9] reports that state-of-the-art large language models (LLMs) undergo pre-training on extensive corpora of unlabeled text to gain a foundational understanding of language [1]. This knowledge can then be fine-tuned on smaller, domain-specific datasets, which is crucial for grooming detection due to the scarcity of labeled datasets stemming from privacy and legal issues [1, 4].

[1] "Throughout this thesis, two terms 'understand' and 'understanding' are used as shorthand for the capacity of LLMs to process and generate text associated with human sense. But this does not imply that LLMs possess human-like cognitive understanding, rather we can say the nature and extent of LLMs' understanding remains as an open question in the field [9]."

Borj et al. [6] illustrated the effectiveness of transfer learning in grooming detection using pre-trained RoBERTa models. Their approach successfully transferred language representations to the context of detecting predatory conversations, resulting in strong classification performance despite limited grooming-specific data. The pre-trained models enhanced vocabulary knowledge and managed typographical errors, a common obstacle in online chat data processing.

Hamm [9] further explored the application of LLMs in grooming detection by fine-tuning the LLaMA 3.2 1B model on the PAN2012 dataset, focusing on grooming conversation detection and predatory author identification. Using parameter-efficient fine-tuning techniques, the adapted LLaMA model achieved remarkable F1 and F0.5 scores of 0.99 in predatory author detection, surpassing both conventional machine learning methods and prior LLM versions [9]. This outcome underscores that modern LLMs can excel in grooming detection tasks, even with fewer parameters, when fine-tuned effectively.

2.3.3 Large Language Model Applications

Recent work has begun exploring LLMs beyond binary grooming detection. Hamm [9] compared traditional machine learning with LLaMA 3.2 1B, focusing on predator tone. By separating conversations into positive and negative sentiment categories, Hamm showed that negative-tone grooming contains subtle patterns harder to distinguish from non-grooming content [9]. This suggests predators use varying strategies that affect detection performance. However, the work remained at binary classification and sentiment analysis, not strategy identification.Other studies have similarly stayed at the binary classification level.

Sallahuddin et al. [3] built SafeCon, a real-time detection system using the Universal Sentence Encoder, PCA, and K-means clustering, but it identified predatory conversations generally rather than by specific strategy. Kim et al. [10] advanced detection for Korean-language grooming using KcELECTRA, yet also operated at binary harmful-versus-safe classification. The field has progressed from keyword matching through classical machine learning to LLMs, improving accuracy at each step [6, 9]. However, detection granularity has not matched model sophistication. Current systems can identify grooming conversations with near-perfect accuracy, but cannot characterize the specific strategies being used within them.

2.4 Research Gaps and Opportunities

Table 2.1 summarizes the four key research gaps identified in the literature. Each gap reflects a limitation in existing work, alongside how this study addresses it. The gaps span model capability, dataset generalization, annotation methodology, and training strategy, collectively motivating the design and scope of this study.

Table 2.1: Research Gaps in Automated Grooming Detection and Corresponding Contributions of the Present Study

#	Research Gap	Current State	Present Study's Contribution
1	Strategy-level classification using LLMs	Taxonomies exist [1, 2] and binary classification performs well [6, 9], but no LLM has been applied to classify specific grooming strategies.	Applies a fine-tuned LLM to classify specific grooming strategies within conversations.
2	Cross-platform generalization	Most studies evaluate only on PAN2012 [13, 9, 14, 5], which has known limitations [5]. Modern platforms show different grooming patterns [4].	Evaluates the fine-tuned model on the LiveMe Grooming Dataset [4] to test generalization.
3	Strategy-level annotated datasets	Existing datasets provide only binary labels. Few stage-annotated datasets cover only broad phases, not specific tactics [1], [5].	Develops a manually annotated dataset with five distinct late-stage grooming strategies, including primary and secondary labels.
4	Impact of manual annotation on LLM performance	Hamm [9] fine-tuned LLaMA on binary PAN2012 labels, but strategy-level human annotation has never been used as a training signal.	Compares fine-tuning on human-annotated strategy labels against a strategy-informed prompt without fine-tuning.

3

Method

This chapter presents the research methodology employed to fine-tune LLM and investigate its effectiveness in identifying specific grooming strategies. The methodology used for this study is a combination of Design Science Research strategy and Experimental Evaluation strategy. It progress through five sequential phases.

3.1 Research Strategy

This study employs a multi-method research strategy that combines Design Science Research (DSR) [15] with quasi-experimental evaluation [13, 16]. DSR provides a rigorous framework for creating and evaluating IT artifacts that address identified real-world problems with a contribution to theoretical knowledge [15]. Following Peffers et al.'s six-step DSR methodology [17] illustrated in Figure 3.1.

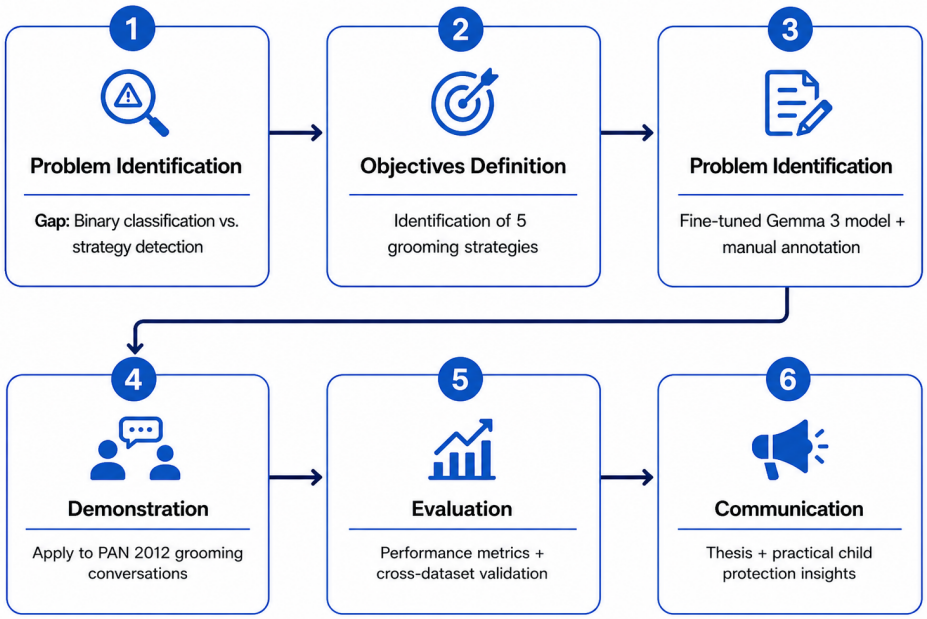


Figure 3.1: Peffers' DSR framework applied to the grooming strategy detection

The evaluation phase employs a quasi-experimental design [18] comparing two conditions to isolate the specific contribution of fine-tuning on manually annotated data.

- **Condition A (strategy-informed/ Prompt based, no fine-tuning)**
  - The unmodified Gemma 3-4B model uses a detailed prompt defining all strategies and indicators, without any training on annotated data.
- **Condition B (fine-tuned with explained strategy and manually annotated data)**
  - The same model is adapted via LoRA and trained on 120 annotated conversations, using an identical strategy definition.

Performance is measured using accuracy, precision, recall, and macro-averaged F1-score computed per strategy class and in aggregate [19]. Macro-averaged F1 is used as the primary summary metric to ensure equal weight across all five strategy classes regardless of their frequency in the annotated dataset. Cross-dataset generalization is assessed by applying the fine-tuned model to the LiveMe Grooming Dataset [4] without further training,

directly addressing the cross-platform generalization gap identified by Borj et al. (2022) [7]. Both conditions are evaluated on the same held-out test set of approximately 40 conversations.

Overall, the study adopts a mixed-methods approach integrating quantitative performance analysis with qualitative manual annotation, combining rigorous numerical evaluation with contextual understanding of model behaviors [18].

3.2 Alternative Research Strategies

Pure Experimental research with fully controlled variables was initially considered. However, it was deemed unsuitable due to the previously conducted studies and the research gaps identified through the literature survey. Grooming detection requires analysis of naturalistic conversation data rather than artificially constructed scenarios. Creating a realistic experimental grooming conversation could raise severe ethical concerns and is more likely to fail in capturing authentic predatory communication patterns. Action research with iterative cycles of active practitioner participation was deemed infeasible within master's thesis constraints and the ethical complexities of involving law enforcement in ongoing research.

The Design Science framework, combined with experimental evaluation, was selected because it accommodates both artifact creation and empirical assessment, aligning with grooming detection studies conducted by Hamm and Vogt et al., and because it provides established evaluation criteria that address an available research gap in the existing studies [8, 9].

3.3 Methodology Overview

Figure 3.2 illustrates a simplified overview of the five-phase research workflow of this study.

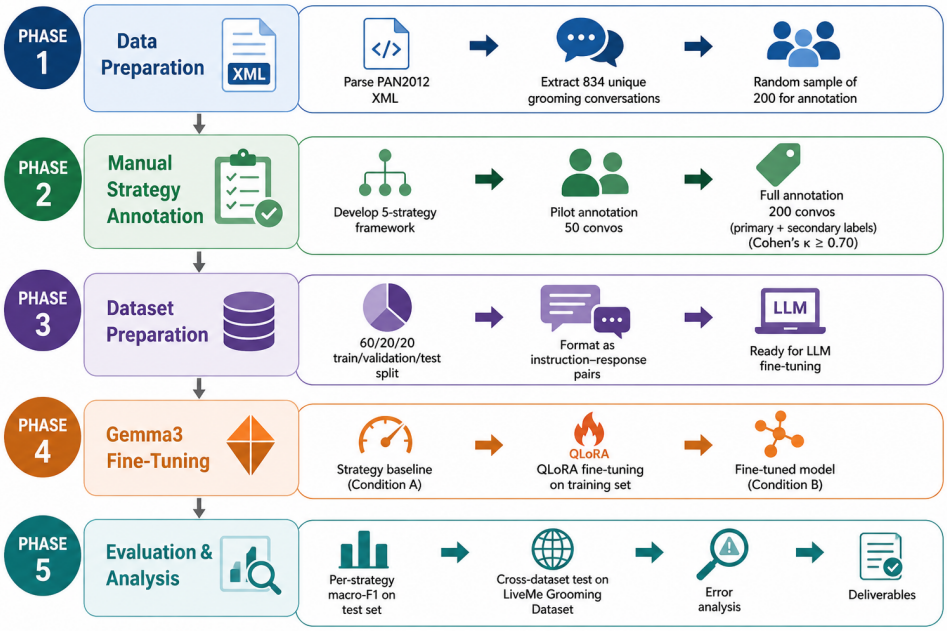


Figure 3.2: Five-phase research methodology overview

3.4 Data Collection

The study employs two primary data sets as inputs. For the first 4 phases of the methodology we used PAN2012 data set which was created for the International Sexual Predator Identification Competition by Inches and Crestani (2012) [3, 8, 9]. For the last phase of the process, we had incorporated The LiveMe Grooming Dataset, derived from Lykousas and Patsakis (2021) [4], provides a substantially different communicative context from PAN2012. The fine-tuned model is applied to this dataset without additional training, testing whether learned strategy representations transfer across platforms with different communication styles. This directly addresses a cross-platform generalization gap identified by Borj et al. (2022) as a major open problem in grooming detection [10].

3.4.1 Alternative Data Collection Methods

Several alternative data collection approaches were available to be considered, but ultimately deemed infeasible to the sensitivity of the data that this thesis aims to analyze. First option was to request authentic grooming conversation records from law enforcement agencies which would provide the most ecologically valid data. However such data would be classified

as evidence in either ongoing or completed criminal investigations and is subjected to strict confidentiality requirements, making access virtually impossible for academic research. Another alternative would be involve researchers posing as minors on selected online platform and collect naturalistic grooming data. This approach could raise several ethical concerns including endangerment of the researchers, the risk of interfering with actual law enforcement operation, and most of all the psychological burden on the researcher. Third option considered was to use another LLM such as ChatGPT to generate synthetic grooming conversations. This approach obviously not going to work, mainly these commercial LLMs would not ideally generate these type of sensitive conversations due to safety guardrails, and also reliance on synthetic data would introduce unknown biases into model training.

Considering all above constraints on alternative data collection methods, the use of existing publicly available research datasets represent the ethically sound and practically feasible approach for this study.

3.5 Sampling Strategy

The PAN2012 test corpus is distributed as a single XML file containing approximately 155,000 conversations. To isolate the grooming-positive subset, a Python extraction script was applied to the corpus in conjunction with the Problem 2 ground truth file, which labels 6,478 individual message lines as predatory. The script parsed the XML, identified all conversations containing at least one ground-truth-labeled grooming message, and extracted them as complete conversation records. This process yielded 834 unique grooming conversations, because multiple labeled messages frequently belong to the same conversation. Each extracted conversation was saved with its full message sequence, author identifiers, and a flag marking which specific lines carried grooming labels. Summary statistics confirmed that the extracted conversations have a median length of 91 messages, involve two persons in 97% of cases, and contain a median of six grooming-labeled messages per conversation, representing approximately 7% of each conversation's total messages. From the 834 extracted conversations, 200 were selected via random sampling to form the manual annotation corpus. This sample was exported in three formats: a structured JSON file retaining all metadata, a CSV annotation sheet with one row per conversation for recording labels, and 200 individual plain-text files allowing annotators to read each conversation in a clean, readable format. The remaining 634

conversations were retained as a reserve pool in case the annotation coverage comes out more biased to a specific category and needs further investigation.

3.6 Data Analysis

As explained above, this research adopts a mixed methods approach to data analysis, integrating qualitative and quantitative techniques to address the research question [18]. The qualitative component consists of systematic manual annotation of grooming conversations by both researchers following a structured six-step framework grounded in established grooming theory[1, 7]. This process produces the labeled dataset that served as both the training signal for model fine-tuning and the ground truth for performance evaluation. The quantitative component consists of fine-tuning an LLM on the annotated data and evaluating performance using standard NLP classification metrics computed per strategy class [7, 19] . Model performance is assessed under two conditions: a strategy-informed prompt baseline with no fine-tuning, and a fine-tuned model trained on the manually annotated conversations, enabling direct measurement of the contribution of the annotated dataset to classification performance [8, 9] . Cross-dataset generalization is further evaluated by applying the fine-tuned model to the LiveMe Grooming Dataset [4] , testing whether learned strategy representations transfer across platforms with different communicative contexts.

3.6.1 Five Grooming Strategies and Their Indicators

The five strategies and their indicators were developed for this study by synthesizing established grooming stage models from O'Connell [1] , the comprehensive behavioral characteristics cataloged by Borj et al. [7] , and the grooming stage features documented by Vogt et al. [9]. Each strategy is operationalization through five indicators. Table 3.1 presents the indicators for each strategy.

Table 3.1: Grooming Strategies and Their Indicators

Strategy	Indicators
Sexual Desensitization [1, 7, 9]	1. Gradual introduction of sexual topics; sexually oriented compliments or body descriptions; 2. Sharing/requesting sexual images, videos, or stories; 3. Questions about sexual experience or preferences; 4. Progressive escalation of explicit language; 5. Normalizing sex discussion as routine between them specifically.
Sexual Normalization [1, 7, 9]	1. Framing age differences or sexual requests as normal; claiming 'everyone does this'; 2. Using guilt to overcome resistance ('you would if you trusted me'); 3. Presenting sexual behavior as a sign of maturity or love; 4. Dismissing legal prohibitions or the victim's concern; 5. Comparing the relationship to adult consensual relationships.
Isolation & Secrecy [1, 7, 9]	1. Explicit instructions not to tell parents or friends; 2. Discouraging the victim from seeking outside help; framing secrecy as protection ('they wouldn't understand'); 3. Warnings about consequences if discovered; 4. Requests to delete messages or move to private channels; 5. 'Us vs them' framing against family.

Strategy	Indicators
Sexual Solicitation & Exploitation [1, 7, 13]	1. Direct requests for explicit photos, videos, or sexual acts on camera; 2. Demands or proposals for offline sexual meetings; 3. Explicit sexual role-play, initiation or continuation; 4. Requests for specific sexual acts, pressure to continue when victim resists; 5. Leveraging previously shared content to demand further compliance.
Control & Coercion [1, 7]	1. Threats (explicit or implicit) if victim refuses or tries to leave; 2. Emotional manipulation to prevent victim ending contact; 3. Aggressive responses when demands are refused using shared secrets or images as implicit; 4. Blackmail; expressions of possessiveness or ownership; 5. Gaslighting; denying or minimising the victim's discomfort.

3.6.2 Manual Annotation Process and Decision Logic

Manual annotation is the process by which each of the 200 sampled PAN2012 grooming conversations is assigned a primary and secondary strategy label drawn from the five-strategy taxonomy described in Section 3.6.1. The annotation framework was developed specifically for this study because no existing labeled dataset distinguishes between the five late-stage grooming strategies targeted here. The framework prioritizes two properties. The first one is reproducibility; the same decision criteria applied by any annotator should produce the same label. The second one is ecological validity, where the criteria should reflect how grooming strategies actually manifest in conversation, not just how they are described in theory.

Here, we consider the full conversation as one Annotation Unit, not the individual lines. Each conversation is treated as a single document

from beginning to end. Labels are assigned to reflect the dominant strategy across the entire conversation arc, rather than the most frequent single message type. This is important because a predator may spend 60 messages on ordinary small talk before the manipulation becomes visible in the final 30 messages. Labeling individual messages would produce a misleading picture in which 'small talk' dominates. Labeling the full conversation captures the predator's overall intent. This conversation-level approach is consistent with how grooming is theorized in the literature. O'Connell (2003) describes grooming as a process that unfolds across time, where early messages establish the conditions for later manipulation [1].

As illustrated in Table 3.2, annotation proceeds through six steps. The first two steps involve reading and interpretation before any scoring occurs. Steps 3 and 4 apply the indicator framework and determine the labels. Steps 5 and 6 verify the decision.

Table 3.2: Annotation Steps and Descriptions

Step	**Description**
Step 1 — Read the Conversation	Identify Opening tone, Turning Point and End Goal
Step 2 — Identify the Conversational Arc	Identify what predator is trying to get from victim
Step 3 — Score Indicators	Count the number of indicators that were identified in Step 2
Step 4 — Primary Label	Arc + highest indicator count
Step 5 — Secondary Strategy	Second highest indicator count
Step 6 — Sanity Check	Victim's responses, density check and arc consistency

Step 1 - Reading the Conversation

Here, the annotator should try to build three mental impressions via reading the whole conversation.

- · **Opening tone: How does the predator first establish contact? Friendly, flattering, neutral?**
- · **Turning point: Where does the conversation shift from neutral to predatory? What triggers it?**
- · **End goal: What is the predator clearly trying to achieve by the final messages?**

The end goal is considered the anchor; the scoring based on the indicators should confirm these impressions, not replace them.

Step 2 - Identifying the Conversational Arc

Before thinking about the number of indicators in a specific strategy category, the annotator must write one sentence answering the question **"By the end of this conversation, what is the predator clearly trying to get the victim to do or accept?"** This arc sentence becomes the primary hypothesis. The scoring in Step 3 either confirms it or overrides it with evidence. Table 3.3 shows an example sentence for each strategy category.

Table 3.3: Primary Labels and Arc Sentences

Likely Primary Label	Arc Sentence
Sexual Desensitization	Escalating sexual language and body descriptions progressively.
Sexual Normalization	Making sexual topics feel normal and discussing them as routine.
Isolation and Secrecy	Telling the victim not to tell anyone.
Sexual Solicitation & Exploitation	Pushing the victim to meet up plans.
Control and Coercion	Threatening or emotionally pressuring the victim when they resist.

Step 3 - Score Indicators

For each strategy, score each of its 5 indicators as present or absent. But a strategy only counts as detected if it clears its minimum signal threshold. The threshold varies depending on how explicitly diagnostic the strategy would be. Table 3.4 below shows the minimum number of messages we consider for one specific strategy, so the important point is that if a strategy does not clear its minimum message threshold, it is not eligible for labeling, regardless of how many indicators it triggered.

Table 3.4: Strategy Thresholds and Reasoning

Strategy	Min. Messages Showing It	Reasoning
Sexual Desensitization	$\geq$2 messages with sexual content escalation	Explicit content is diagnostically strong; 2 occurrences establish a pattern of escalation.
Sexual Normalization	$\geq$2 messages reframing or applying guilt	Single guilt comment can be conversational; 2+ shows a deliberate reframing effort.
Isolation & Secrecy	$\geq$3 messages with secrecy enforcement	Secrecy requests are common in ordinary teen friendships; needs a clear repeated pattern to be diagnostic.
Sexual Solicitation	$\geq$1 direct explicit request	A single direct request for explicit videos, images, or a meeting is inherently diagnostic.
Control & Coercion	$\geq$1 threat, $\geq$2 emotional pressure messages	Explicit threats are unambiguous at 1; emotional pressure needs 2 to rule out normal frustration.

When scoring is completed, and there are more than 2 strategies with the highest scores, a problem arises when moving into Steps 3 and 4. So a tie-breaking rule is needed to resolve the overlaps by prioritizing grooming behaviors. Figure 3.3 illustrates a hierarchical tiebreaker system where grooming behaviors are ranked from most to least severe, with sexual desensitization as the strongest indicator and isolation as a supporting signal, guiding decisions by prioritizing the highest-level behavior present.

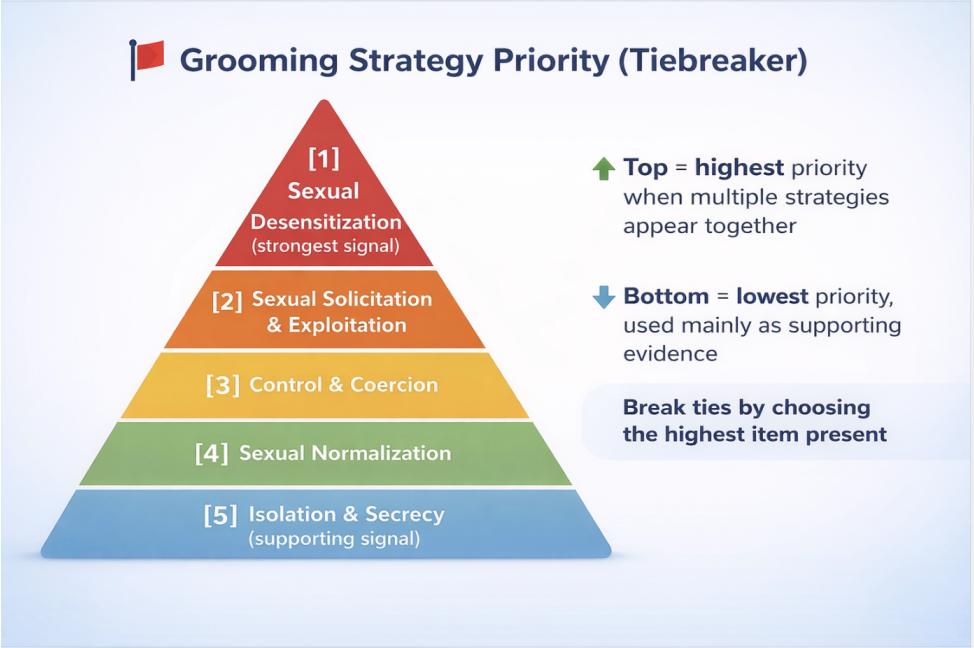


Figure 3.3: Tie Breaker: Priority List

Step 4 and Step 5 - Assigning Primary and Secondary Labels

Among the strategies that passed in step 3 considered for primary labeling. This step is also assigned using two criteria, as explained below.

Criteria A: Conversational arc

If the arc sentence from Step 2 points to one of the eligible strategies, then that strategy is the primary label regardless of how many indicators the other eligible strategies may have triggered.

Criteria B: Indicator count

If the arc is unclear or ambiguous, the eligible strategy with the most indicators present becomes the primary label.

So in a conversation, when both A and B agree, the decision is made with high confidence. When they disagree, Criteria A should take priority. Table 3.5 displays a few situational examples with a final decision with proper reasoning.

Table 3.5: Labeling Decision Rules and Reasoning

Situation	Decision	Reasoning
A and B agree	Arc strategy = primary. Assign the supporting strategy as secondary.	High confidence.
A is clear but B has higher score in different category	Trust the arc. The strategy the predator is building toward is primary; the strategy with more indicators is secondary.	Arc reveals intent. Count reveals the tool used to get there.
A is not clear	Highest indicator count = primary. Second highest = secondary.	Fallback when conversational intent is genuinely ambiguous.
Tie in indicator count	Apply tiebreaker priority: Desensitization > Solicitation > Coercion > Normalization > Isolation.	Priority order reflects diagnostic severity of each strategy. Tiebreaker is explained in Section 3.6.2, Figure 3.3.
Only one strategy eligible	That strategy = primary. Secondary = None.	Recorded explicitly.
No strategy eligible	Both labels = None / General.	Conversation is confirmed predatory by ground truth, but the strategy is not among the selected list.

Table 3.6 illustrates how the annotation framework handles different conversation patterns, including different situations explained in Table 5.

Table 3.6: Worked Examples of Labelling Decision Logic

Conversation Pattern	**Indicator Scores**	**Decision Logic**	**Labels Assigned**
Conversation includes a direct meet request. Prior or in-between messages used normalization to lower resistance.	Normalization hits 4/5 indicators. Solicitation hits 1/5 (the request itself).	Arc = Solicitation. Arc overrides count.	Primary: Sexual Solicitation Secondary: Sexual Normalization
Predator progressively escalates body descriptions across 12 messages. Victim expresses discomfort repeatedly.	Desensitization hits 4/5 indicators. Normalization hits 2/5.	Arc = Desensitization. Count agrees.	Primary: Sexual Desensitization Secondary: Sexual Normalization
Conversation ends with explicit threat after victim refuses to send images.	Desensitization hits 3/5 earlier. Coercion hits 2/5 at the end.	Arc = Coercion (endpoint). Arc overrides higher earlier count.	Primary: Control & Coercion Secondary: Sexual Desensitization
One mention of 'don't tell your parents' at the end. No other indicators.	Isolation hits 1/5 but does not clear the $\geq$3 message threshold.	Isolation ineligible. No other strategy clears threshold.	Primary: None / General Secondary: None

Step 6 - Sanity Check and Recording Output

Before finalizing, ensure the strategy is validated by the victim's response and by its continuation throughout the conversation, rather than as a brief spike. The final annotation output for each conversation is a structure.

The final annotation output for each conversation is a structured record containing six fields: conversation number, conversation identifier, the primary strategy label, the secondary strategy label, the indicator score vector, and the arc sentence annotator note. This structured output serves as both the training dataset for Gemma 3 fine-tuning and the ground truth for evaluation.

3.6.3 Inter-Annotator Reliability

Inter-annotator reliability was assessed to ensure the consistency and credibility of the manual annotation process. Since conducting a dedicated pilot annotation phase was not feasible within the time constraints of this study, reliability was instead established through a structured calibration session held before full annotation began. In this session, both annotators jointly reviewed the annotation framework, discussed the six-step procedure, and worked through a set of example conversations together to align their understanding of each strategy's indicators and detection thresholds. This approach to reliability establishment through framework calibration and shared understanding before independent annotation is recognized as a valid alternative to pilot-based reliability testing in time-constrained research settings [6, 13].

Following the calibration session, both annotators independently annotated the full set of 200 PAN2012 conversations and 40 LiveMe conversations. Inter-annotator agreement was then measured using Cohen's kappa ($\varkappa$), and the final kappa score achieved across both annotators is reported in the Results chapter.

3.6.4 Mode Selection and Fine-Tuning Gemma 3

The second section of the data analysis process involves fine-tuning the Gemma 3 LLM on the manually annotated dataset produced in Section 3.6.2. The objective is to analyze whether a model trained on human-annotated examples of grooming strategies can classify those strategies more accurately than a model operating on strategy knowledge alone, which directly addresses the research question.

Computational Infrastructure.

Fine-tuning is conducted on a remote server accessed via SSH. Table 3.7 summarizes the full hardware specification.

Table 3.7: Hardware and System Specifications

Component	Specification
CPU	AMD Ryzen 9 9950X — 16 cores, 32 threads, up to 5.76 GHz; AVX-512 support
System RAM	60 GB DDR5 (56 GB available); 8 GB swap
GPU	NVIDIA GeForce RTX 4090 — 24 GB GDDR6X VRAM
CUDA Version	13.1 — full support for bfloat16 precision and Flash Attention 2
GPU VRAM Usage	∼187 MB used at rest — effectively 24 GB available for training
Access	Remote server via SSH

Justification for Gemma 3

Gemma 3 [11, 12] is selected as the base model for this study for four reasons. First, it is openly available under a permissive research license with full reproducibility without API access restrictions. Second, the Gemma 3-4B parameter variant balances model capacity against computational feasibility: it is large enough to capture complex linguistic patterns in grooming conversations, yet small enough to fine-tune with full bfloat16, no quantization (LoRA) on a single GPU within academic computing constraints. Third, Gemma 3 is built on the transformer architecture with multi-head self-attention [11, 12], enabling it to model long-range dependencies across entire conversations, which would be beneficial in grooming strategy identification. Fourth, Gemma 3 has not previously been applied to strategy-level grooming detection, making this study a novel application of the model in this domain.

The choice of Gemma 3-4B over LLaMA 3.2 1B (the model used by Hamm, 2025 [14]) is deliberate. LLaMA 3.2 1B is optimized for edge deployment and sacrifices model capacity for speed. For a five-class strategy labeling task, where the model must distinguish between overlapping manipulation tactics. Gemma 3-4B provides approximately four times the parameters

of LLaMA 3.2 1B while remaining efficiently fine-tunable under academic hardware constraints through LoRA[20].

3.6.5 Evaluation Design

The evaluation follows a quasi-experimental design comparing two conditions that differ on one variable, which is the presence or absence of fine-tuning on manually annotated data while holding the base model, strategy definition, and evaluation setup constant [13]. Table 3.8 summarizes the design.

Table 3.8: Comparison of Condition A and Condition B

Element	**Condition A — Strategy Baseline**	**Condition B — Fine-Tuned Model**
Base Model	Gemma 3-4B instruction-tuned	Gemma 3-4B instruction-tuned
Strategy Definitions	Provided in system prompt	Provided in training prompt
Labelled Training Data	None	200 annotated conversations
Weight Updates	No	Yes — via LoRA on RTX 4090
Represents	Best possible prompt baseline	Fine-tuned artifact (this study)

3.6.6 Fine-Tuning Procedure and Configuration

Fine-tuning is performed using Low-Rank Adaptation (LoRA), a parameter-efficient method that introduces small trainable rank-decomposition matrices into the transformer attention layers while keeping the pre-trained weights frozen [20]. The procedure, in combination with the full bfloat16, no quantization (LoRA), can be executed on a single academic GPU. Implementation used the Hugging Face Transformers and PEFT libraries, with Flash Attention 2 enabled for accelerated training.

The training examples are in the form of instruction-response pairs. The instruction component contains the system prompt with strategy definitions and indicator descriptions. The input component contains the full conversation text up to 512 tokens. The response component contains the

six-column annotation output, the same as in the manual annotation. Intention of this step was to directly mirror the structured six-step thinking process of the human annotators.

The table 3.9 presents the complete hyperparameter configuration, with settings optimized for the RTX 4090 hardware.

Table 3.9: Fine-Tuning Hyperparameter and Configuration

Parameter	**Setting**
Base Model	Gemma 3-4B (instruction-tuned checkpoint, `google/gemma-3-4b-it`)
Precision	bfloat16 — native GPU precision on RTX 4090; no quantization required
Fine-Tuning Method	LoRA — rank $r = 16$, alpha$= 32$, applied to all attention layers (q, k, v, o projections)
Training Epochs	5–10 with early stopping; patience $= 2$ validation epochs
Learning Rate	2×10^{-4} with cosine annealing decay schedule
Batch Size	16 per GPU step; gradient accumulation steps $= 2$ (effective batch $= 32$)
Max Sequence Length	512 tokens — RTX 4090 VRAM comfortably supports full-length sequences
Optimizer	AdamW; weight decay $= 0.01$; gradient clipping $= 1.0$
Attention	Flash Attention 2 — enabled via CUDA 13.1 for faster attention computation
GPU VRAM Requirement	$\sim$14–16 GB during training — within 24 GB RTX 4090 budget
Estimated Time	15–30 minutes per epoch — substantially faster than CPU-based approaches

Following annotation, the 200 labeled conversations are partitioned into training, validation, and test sets using a 60/20/20 split. Table 3.10 shows the split proportions and the purpose of each partition.

Table 3.10: Dataset Split Configuration

Split	Proportion	Approx. n (of 200)	Purpose
Training	60%	∼120	Fine-tune Gemma 3 LoRA adapter weights
Validation	20%	∼40	Hyperparameter tuning; early stopping criterion
Test	20%	∼40	Final evaluation: held out until all training is complete

Evaluation and Cross-Dataset Generalization

Both Condition A and Condition B are evaluated on the same held-out test set of 40 PAN2012 conversations using accuracy, precision, recall, and macro-averaged F1-score computed per strategy class [7, 19]. Macro-averaged F1 is the primary summary metric because it weights all five strategy classes equally, regardless of their frequency in the test set, preventing dominant classes from masking poor performance on rarer strategies such as Isolation and Secrecy or Control and Coercion.

Cross-dataset generalization is assessed by applying the fine-tuned Condition B model to the 40 manually annotated LiveMe Grooming Dataset conversations [13] without further training, testing whether learned strategy representations transfer across platforms with substantially different communicative contexts [7]. Performance on the LiveMe set is reported using the same four metrics, and the gap between in-domain and cross-dataset performance is interpreted as a measure of model robustness. All results are reported in the Results chapter.

3.7 Ethical Considerations

Ethical aspects of the research should be discussed explicitly, as the studied topic is rather sensitive. All of the data utilized in this study were anonymized discussions based on existing, publicly available studies [13, 11]. There is no information gathered with minors or on active platforms

or law enforcement. There were no participant consent issues, as no new data were gathered involving human subjects. This study explicitly does not involve the application of detection models to live systems or in real-time tracking of active communications [6, 13]. The research offers a proof of concept and performance metrics alone. But in practice, further ethical guidelines, law, and privacy impact assessment would be necessary to implement it in the real world.

Human annotators are exposed to explicit and psychologically disturbing material during the process of annotation. Annotators were also told up front about the kind of content they were working with, and they are not expected to annotate more than they feel comfortable with, and they are encouraged to have regular breaks. Annotators have a risk of secondary traumatic stress, which is recognized, and annotators were advised to seek support resources when necessary.

The results of the research, the annotation framework, the refined model, and benchmarks will be used in child protection. Nonetheless, any automated detection system has risks of false positives (marking innocent conversations) and false negatives (not detecting real grooming). False negatives in the child protection context have a higher potential for harm compared to false positives. The assessment system is specified to report explicit precision and recall per class, and the results discussion is also a subject of the ethical implications of the observed error distributions.

4

Results

This chapter presents the empirical results of the study in three sections. The first section reports the outcomes of the manual annotation, including inter-annotator reliability statistics. The second section presents the classification findings for Condition A and Condition B in Gemma 3. The final section summarizes the key results and their implications for the research question.

4.1 Data Collection and Analysis

4.1.1 PAN2012 Dataset Extraction

The PAN2012 corpus was processed using a Python extraction script applied in conjunction with the ground truth-problem 2 file, yielding 834 grooming conversations [9]. From these, 200 were randomly sampled for annotation. Summary statistics are in Table 4.1.

Statistic	**Value**
Total conversations extracted	834
Sampled for annotation	200
Reserve pool	634
Median conversation length (messages)	91
Two-participant conversations	97%
Median grooming-labelled messages per conversation	6 ($\sim$7% of messages)

Table 4.1: Summary statistics of extracted PAN2012 grooming conversation

4.1.2 LiveMe Dataset Extraction

The LiveMe Grooming Dataset originates from Lykousas and Patsakis (2021) and is distributed as raw chats [4]. Unlike PAN2012, this dataset did not provide a pre-labeled grooming conversation; it contained a large pool of chat logs from where grooming-relevant conversations must be identified and extracted. A keyword-based Python Script was developed to identify grooming-risk conversations from the first 50,000 conversation IDs. This script filtered two participants' conversations, then scored each using a weighted keyword dictionary spanning eight grooming-related categories related to this study, and then these conversations were ranked with weights of 1-4 reflecting diagnostic severity. Conversations were ranked by score, and the top 100 conversations were exported as anonymized plaintext files with user identifiers replaced by truncated SHA-1 hashes. From these 100, the top 40 conversations were selected for manual annotation to form the cross-platform generalization test set.

4.1.3 Manual Annotation-PAN2012

Based on the described six steps annotation process in Chapter 3, both annotators independently annotated 200 sampled PAN2012 conversations. Primary and secondary strategy labels were assigned for each conversation following the arc sentence principle. Appendix A includes the annotation results for each annotator and the final agreement between Primary and Secondary Annotation Labels and the Final Arc Sentence.

The inter-annotator reliability was assessed using JASP (version 0.96). Agreement between the two annotators was evaluated using Cohen's kappa, Fleiss' kappa, and Krippendorff's alpha across all five strategy classes separately for primary strategy and secondary strategy. A summary of results is reported in Tables 4.2 and 4.3

Metric	**Overall κ / α**	**SE**	**95% CI Lower**	**95% CI Upper**
Cohen's κ (Annotator 1 – Annotator 2)	0.874	0.029	0.817	0.931
Fleiss' κ (Overall)	0.874	0.044	0.789	0.959
Krippendorff's α (Nominal)	0.874	0.028	0.817	0.926

Table 4.2: Inter-annotator agreement statistics for Primary Strategy Annotation

For the Primary strategy alone, all three agreement measures converged at $\varkappa$ = 0.874, substantially exceeding the minimum acceptable threshold of $\varkappa \geq 0.70$ specified in the methodology. Per-class Fleiss' kappa values

ranged from 0.856 (Sexual Desensitization and Sexual Normalization) to 0.922 (Control and Coercion), indicating consistently strong agreement across all five strategy categories. This level of agreement confirms that the annotation framework produces reliable and reproducible labels suitable for use as ground truth in model training and evaluation.

Metric	Overall κ / α	SE	95% CI Lower	95% CI Upper
Cohen's κ (Annotator 1 – Annotator 2)	0.834	0.032	0.772	0.896
Fleiss' κ (Overall)	0.834	0.040	0.755	0.913
Krippendorff's α (Nominal)	0.835	0.032	0.766	0.895

Table 4.3: Inter-annotator agreement statistics for Secondary Strategy Annotation

Secondary strategy annotation achieved Cohen's kappa of $\varkappa = 0.834$ (SE = 0.032, 95% CI: 0.772–0.896), also exceeding the 0.70 threshold. This result is expected to be slightly lower than primary annotation because secondary labels involve more ambiguous cases where multiple strategies co-occur. Full statistics are presented in Appendix B.

4.1.4 Manual Annotation - LiveMe Dataset

The same annotation process was applied for forty conversations from the LiveMe Grooming Dataset. Inter-annotator agreement for the LiveMe primary strategy annotation achieved Cohen's kappa of $\varkappa = 0.787$ (SE = 0.076, 95% CI: 0.638–0.937), and secondary strategy annotation achieved $\varkappa = 0.772$ (SE = 0.082, 95% CI: 0.611–0.933). Both values exceed the 0.70 threshold, confirming that the annotation framework generalizes reliably to the LiveMe platform context. Agreement statistics are summarized in Table 4.4. The data sheet used for calculation is included as Appendix C, and the Full statistics are presented in Appendix D.

Annotation	Metric	κ / α	SE	95% CI Lower	95% CI Upper
Primary Strategy	Cohen's κ	0.787	0.076	0.638	0.937
	Krippendorff's α	0.789	0.081	0.614	0.927
Secondary Strategy	Cohen's κ	0.772	0.082	0.611	0.933
	Krippendorff's α	0.774	0.085	0.590	0.925

Table 4.4: Inter-annotator agreement for strategy annotations

The slightly lower agreement on the LiveMe dataset compared to PAN2012 ($\varkappa$=0.787 vs. $\varkappa$=0.874) is expected due to its different communicative styles, platform-specific language patterns, and shorter average message lengths.

4.1.5 Dataset Split

The 200 PAN2012 conversations were partitioned before any training began, and the 40 LiveMe conversations also being separately attached to the platform test. This partitioning is summarized in Table 4.5.

Table 4.5: Dataset Splits and Their Purposes

Split	Proportion	Conversations (n)	Purpose
Training	60%	120	Fine-tune Gemma 3-4B LoRA (Condition A)
Validation	20%	40	Monitor training loss and early stopping
Test	20%	40	Final evaluation — both Condition A and Condition B
Cross-platform test (LiveMe)	—	40	Generalisation test — Condition B only

4.2 Findings

4.2.1 Primary Evaluation - PAN2012 Test Set

This section reports classification results for Condition A and Condition B on the 40 randomly selected PAN2012 test conversations. Both conditions use identical system prompts, test data, and evaluation procedures. The sole independent variable is the presence of LoRA fine-tuning in Condition B.

Condition A - Strategy-Informed Prompt Baseline

Condition A evaluates the unmodified Gemma 3-4B instruction-tuned model using the full strategy-informed system prompt but without any fine-tuning on annotated data. Results are presented in Table 4.6.

Table 4.6: Condition A per-class evaluation results)

Strategy	Precision	Recall	F1	Support
Sexual Desensitization	0.3750	0.2143	0.2727	14
Sexual Normalization	0.0000	0.0000	0.0000	4
Isolation and Secrecy	0.0000	0.0000	0.0000	1
Sexual Solicitation and Exploitation	0.7500	0.6316	0.6857	19
Control and Coercion	0.0625	0.5000	0.1111	2
Macro Average	**0.2375**	**0.2692**	**0.2139**	**40**

Condition A achieved an overall accuracy of 40.0% and a macro-averaged F1 of 0.2139. Sexual Solicitation and Exploitation achieved the strongest

result (F1=0.686), attributable to its explicit and unambiguous surface signals. Sexual Normalization, and Isolation and Secrecy both scored F1=0.000. Control and Coercion was predicted 16 times against only 2 manual instances, indicating the model conflates general negative tone with coercive strategy. Full per-conversation results and confusion matrix are in Appendix E.

Condition B - Fine-Tuned Gemma 3-4B

Condition B evaluates Gemma 3-4B after LoRA fine-tuning on 120 manually annotated PAN2012 training conversations. The identical system prompt, test set, and evaluation procedure as Condition A were used. Results are in Table 4.7.

Condition B achieved an overall accuracy of 52.5% and a macro-averaged F1 of 0.2563, representing improvements of +12.5 percentage points in accuracy and +0.042 in macro F1 over Condition A. The most notable finding is the emergence of Sexual Normalization detection, the baseline completely failed to identify this strategy (F1=0.000), while the fine-tuned model achieved F1=0.400 with perfect precision (1.000), indicating that all predictions made by the model were correct. Sexual Desensitization F1 nearly doubled from 0.273 to 0.519. The systematic over-prediction observed in Condition A was eliminated, though this came at the cost of Control and Coercion recall dropping to zero, attributable to the very small number of Control and Coercion test instances (n=2). Isolation and Secrecy remained at F1=0.000 under both conditions, reflecting insufficient test set representation (n=1) rather than a systematic model failure. Full results in Appendix F.

Table 4.7: Condition B per-class evaluation results

Strategy	**Precision**	**Recall**	**F1**	**Support**
Sexual Desensitization	0.5385	0.5000	0.5185	14
Sexual Normalization	1.0000	0.2500	0.4000	4
Isolation and Secrecy	0.0000	0.0000	0.0000	1
Sexual Solicitation and Exploitation	0.5652	0.6842	0.6190	19
Control and Coercion	0.0000	0.0000	0.0000	2
Macro Average	**0.3506**	**0.2390**	**0.2563**	**40**

Additional Fine-Tuning - Ablation Findings

An additional Condition B run using all 160 available conversations produced a macro-averaged F1 of 0.2311 and accuracy of 40.0%, both lower than the 120-conversation model (F1=0.256, accuracy=52.5%). This confirms that the randomly sampled 120 conversations provided a more balanced training signal than the full 160-conversation set. Full ablation results are in Appendix G.

Supplementary Comparison - Gemma 3-12B

As an additional expansion to the original methodology, Gemma 3-12B was evaluated under the same Condition A and Condition B protocols on the same 40 PAN2012 test conversations.

In Gemma 3 model terms "4B" and "12B" refer to the number of trainable parameters in the model. So, Gemma 3-4B contains approximately 4 billion parameters, while Gemma 3-12B contains approximately 12 billion. Larger the parameter count means greater capacity for learning complex patterns, but integrated with requirement of a large computational resources for training and outputs.

Table 4.8 presents a four-way comparison across both model sizes and conditions. Gemma 3-12B was run with 4-bit quantization (QLoRA) due to VRAM constraints, which introduces a confounding variable not present in 4B evaluation. Full results are in Appendix H.

Table 4.8: Supplementary four-way Comparison Across Gemma 3 4B and 12B under same Conditions

Strategy	**4B-A**	**4B-B**	**12B-A**	**12B-B**	**Best F1**
Sexual Desensitization	0.2727	0.5185	0.2500	0.2222	4B Cond B (0.519)
Sexual Normalization	0.0000	0.4000	0.5000	0.5000	12B Cond A/B (0.500)
Isolation and Secrecy	0.0000	0.0000	0.6667	0.6667	12B Cond A/B (0.667)
Sexual Solicitation & Exploitation	0.6857	0.6190	0.7368	0.7143	12B Cond A (0.737)
Control and Coercion	0.1111	0.0000	0.2857	0.2857	12B Cond A/B (0.286)
Overall Accuracy	**40.0%**	**52.5%**	**62.5%**	**60.0%**	—
Macro F1	**0.2139**	**0.2563**	**0.4065**	**0.3981**	—

Three observations is clearly visible with this comparison.

1. The 12B baseline substantially outperforms the 4B baseline (macro F1: 0.407 vs 0.214), consistent with larger model capacity.

2. Fine-tuning improved the 4B model (+0.042) but slightly degraded the 12B model (-0.008)

3. For Sexual Normalization, the 12B baseline already achieves F1=0.500 without fine-tuning, suggesting sufficient model capacity can detect this strategy from definitions alone.

4.2.2 Cross-Platform Generalization - LiveMe Test Data Set

This section reports how the fine-tuned Condition B model generalized into 40 LiveMe conversations without any further training. The summarized results are shown in Table 4.9, where a comparison is made between the Gemma 3-4B fine-tuned model and the Gemma 3-12B fine-tuned model. Full results in Appendix I.

Table 4.9: Condition B - Cross-Platform Evaluation (LiveMe Dataset)

Metric	**Gemma 3-4B**	**Gemma 3-12B**
Overall Performance		
Overall Accuracy	43.6%	53.8%
Macro F1	0.3367	0.4351
Weighted F1	0.4479	0.5306
Per-class F1 Scores		
Sexual Desensitization (SD)	0.4000	0.4444
Sexual Normalization (SN)	0.0000	0.0000
Isolation and Secrecy (IS)	0.5333	0.5882
Sexual Solicitation and Exploitation (SSE)	0.5000	0.5714
Control and Coercion (CC)	0.2500	0.5714
Precision / Recall		
Macro Precision	0.4561	0.5356
Macro Recall	0.3399	0.5081

The Gemma 3-4B model achieved an overall accuracy of 43.6% and a macro F1 score of 0.3367 on the LiveMe dataset under Condition B. At the class level, performance was uneven, with Isolation and Secrecy yielding the strongest result (F1 = 0.5333) and Sexual Solicitation and Exploitation following closely (F1 = 0.5000). Control and Coercion was notably weaker (F1 = 0.2500), while Sexual Normalization received a score of 0.0000 due to its absence from the evaluation set.

The Gemma 3-12B model demonstrated consistently stronger performance, achieving an overall accuracy of 53.8% and a macro F1 of 0.4351. Class-level results were more balanced across categories, with Control and

Coercion showing the most substantial improvement over the smaller model (F1 = 0.5714), and both Isolation and Secrecy (F1 = 0.5882) and Sexual Solicitation and Exploitation (F1 = 0.5714) reaching comparable scores. Sexual Normalization remained undetected, again reflecting its absence from the evaluation set.

Comparing the two models, scaling from 4B to 12B parameters produced consistent gains across all reported metrics, with overall accuracy improving by 10.2% and macro F1 increasing from 0.3367 to 0.4351. The most pronounced improvement was observed in Control and Coercion, where the 12B model more than doubled the F1 score of its smaller counterpart, suggesting that increased model capacity aids in detecting strategies characterized by subtler or more implicit linguistic cues.

4.3 Summary

Key findings can be summarized into mainly 5 points.

- Inter-annotator reliability was strong across both the PAN2012 and LiveMe datasets, with agreement levels exceeding the 0.70 threshold. This indicates that the proposed annotation framework was applied consistently across both datasets.
- Fine-tuning the Gemma 3-4B model produced meaningful improvements on the PAN2012 dataset. Condition B outperformed Condition A, with accuracy increasing by 12.5% and macro F1 improving by 0.042. Most notably, the Sexual Normalization category improved from an F1 score of 0.000 to 0.400. This suggests that exposure to labeled examples enabled the model to learn strategy-specific patterns that prompt definitions alone could not capture.
- The results also indicate that training data quality may be more important than training data quantity. The model trained on 160 conversations performed slightly worse than the model trained on 120 conversations, with macro F1 scores of 0.231 and 0.256 respectively. This suggests that adding more training instances does not necessarily improve performance when additional data may introduce greater variability or annotation noise.
- The supplementary 12B comparison highlights the influence of both model size and quantization. While the 12B baseline outperformed

the 4B baseline, fine-tuning led to a decline in 12B performance under the 4-bit quantised setting. This supports the methodological decision to use full bfloat16 precision for the primary 4B evaluation.

- Cross-platform generalization was also encouraging. Both fine-tuned models achieved meaningful performance on the LiveMe dataset without further training.

Overall, the research question can be answered positively. The findings show that fine-tuned large language models can identify specific grooming strategies more effectively than strategy-informed prompting alone. In addition, the learned strategy representations demonstrated the ability to transfer across platforms [1].

[1] Source code and annotated datasets are available from authors upon request.

5

Discussion

This chapter interprets the results presented in Chapter 4 and considers possible ways that future research could overcome the identified limitations. The chapter begins with a consideration of the results and the potential impact of methodological delimitation on them. The chapter concludes with a synthesis of the key findings to answer the research question posed in Section 1.2.

5.1 Analysis of the Results

5.1.1 Analysis of Result Findings

1. **Fine-tuning improved strategy-level classification performance.**
    a. As per the goal of the study, results demonstrated that fine-tuning on manually annotated grooming conversations produced a clear improvement over prompt-based inference alone.
    b. F1 score values remain below the near-perfect binary detection results reported by Hamm (F1 = 0.99) [9] and Borj et al. (F0.5 = 0.98) [6], but direct comparison is not appropriate because prior work addressed binary classification, whereas this study required discrimination among five overlapping grooming strategies within the same conversation. This approach has not been attempted in prior research.
2. **Strategy Identification comparison showed varied improvements across each category.**

a. Sexual Solicitation and Exploitation remained the strongest performing strategy class in both conditions. This is expected because the strategy contains explicit surface signals, such as direct requests for meetings or images [1, 2], which are relatively easy to identify even without fine-tuning. The slight decline under Condition B indicates a trade-off in which fine-tuning distributed predictions more evenly across strategy classes rather than over-predicting the most explicit category.

b. Sexual Desensitization showed substantial improvement after fine-tuning. This suggests that the fine-tune model learned escalation trajectories rather than merely detecting sexual vocabulary.

c. Sexual Normalization emerged only after fine-tuning. This finding supports the argument of Borj et al. [6] that grooming signals operate primarily at the level of sentence semantics rather than isolated lexical features.

d. Rare strategy classes, such as Isolation and Secrecy, remained difficult to evaluate reliably in the PAN2012 test dataset due to a lack of annotated data. However, they were more clearly detectable in the LiveMe dataset, which suggests that secrecy-related behaviors may be more explicit on live-streaming platforms.

3. **Training data quantity vs. quality was challenged via the Annex I findings.** The additional 40 conversations may have introduced label variability or annotation noise, which may have caused the overall reduction in accuracy and F1 scores.

4. **Supplementary comparison with the Gemma 3-12B model** showcased overall improvement in accuracy in both conditions compared to the 4B model. However, when comparing both conditions within the same model, fine-tuning reduced the overall accuracy and F1 score slightly. This can be reasonably justified due to the use of 4-bit quantization for the 12B model, required due to VRAM constraints on the GPU.

5. **The cross-platform evaluation on the LiveMe dataset produced encouraging results.** These results indicate that strategy representations learned from PAN2012 can be successfully transferred to a substantially different communicative environment.

5.1.2 Limitations

Several limitations must be acknowledged when interpreting these findings.

The most significant limitation is the small size of the annotated dataset. With only 200 PAN2012 conversations and 40 LiveMe conversations, the per-class sample sizes for rarer strategies such as Control and Coercion and Isolation and Secrecy are insufficient for robust evaluation. This constraint is inherent due to the scope of a master's thesis and the labor-intensive nature of manual annotation on sensitive content. Expanding the dataset would require either additional annotators or longer annotation periods, both of which were infeasible within the available time frame.

Second, the PAN2012 dataset itself has ecological limitations. The conversations were generated through sting operations in which adult volunteers posed as minors [1, 2]. The resulting chat dynamics may differ from authentic predator–victim interactions in terms of vocabulary, emotional responses, and conversational flow. Using this as training data may therefore limit the model's ability to detect strategies as they manifest in genuine grooming scenarios.

Third, the maximum sequence length of 512 tokens truncates longer conversations, which may result in the loss of critical strategy-relevant content. This truncation may disproportionately affect strategies that manifest late in conversations, such as Control and Coercion, which Hamm [9] noted tends to appear at conversation endpoints.

Fourth, the cross-platform evaluation on LiveMe, while producing encouraging results, is limited to 40 conversations that were selected through keyword-based filtering rather than exhaustive annotation of a random sample. This selection method may introduce sampling bias toward conversations with more explicit grooming signals, potentially overstating the model's generalization capability on the broader LiveMe corpus.

Finally, the study evaluates only one model family, Gemma 3 . While the choice of Gemma 3 was justified in Section 3.6.4, the findings may not generalize to other LLM architectures.

5.2 Ethical and Societal Consequences of the Results

This section examines the ethical and societal implications of the findings reported in Chapter 4 and considers the factors that must be addressed when translating these results into real-world applications.

The fine-tuned Gemma 3-4B model achieved only an accuracy of 52.5%, meaning nearly half of all test conversations were misclassified. In the real world, if this system were implemented directly, a substantial proportion of grooming strategies would either go undetected or be incorrectly characterized. As highlighted in Section 3.7, false negatives carry greater potential for harm than false positives in child protection, because a missed grooming conversation may leave a child exposed to continued exploitation [1, 5]. Furthermore, inability to detect one category, Isolation and Secrecy and Control and Coercion, recall dropped to zero after fine-tuning, creates a progressive escalation of the victim's risk [2], and a system that fails to detect them could create a false sense of security among moderators or parents who rely on automated tools.

On the other hand, there are positive aspects to be noted. The emergence of Sexual Normalization detection (F1 rising from 0.000 to 0.400) has meaningful societal value. Identifying how a predator exploits a victim, not just that grooming is occurring, could enable strategy-aware intervention by child protection professionals [2]. The cross-platform transfer to LiveMe further suggests that learned representations are not dataset-specific, which is encouraging for broader applicability. However, the current performance levels are insufficient for unsupervised deployment on live platforms without human-in-the-loop review [3, 5].

Finally, as we acknowledge the requirement to increase the amount of the annotated database, the annotation effort for future work must be accompanied by institutional support for annotator well-being, given the psychologically disturbing nature of the material. Any future integration into live moderation systems would also require balancing child safety with user privacy, with transparent communication about the limitations of automated monitoring [3, 5]. These considerations are beyond the scope of this proof-of-concept study but are essential for a responsible translation of these findings into practice.

5.3 Future Research

The findings of this study open several avenues for future research. The most immediate priority is scaling the annotated dataset. A larger corpus of strategy-labeled conversations, ideally involving multiple annotator teams and diverse data sources, would enable more reliable per-class evaluation and support the training of larger models at full precision. Future studies could explore active learning approaches, in which model predictions are

used to prioritize conversations for human annotation, thereby maximizing the information gained from each annotated example.

Cross-linguistic extension is another valuable direction. The annotation framework and fine-tuning approach developed here are language-independent in principle. Adapting them to other languages, following the precedent set by Kim et al. [10] for Korean, would address the English-language limitation of this study and broaden the practical applicability of strategy-level detection to global platforms.

Finally, real-time deployment represents the ultimate practical goal for grooming detection research. The current study evaluates models in an offline, post-hoc setting. Integrating strategy-level classification into real-time monitoring systems, similar to the SafeCon architecture proposed by Sallahuddin et al. [3], would require addressing latency constraints, false positive management, and the ethical implications of automated surveillance. These operational challenges were deliberately excluded from the present study but should be addressed in future work to translate the proof-of-concept findings into practical child protection tools.

5.4 Use of AI Tools

The use of LLM, as the object of study, was inherent in this research as the core artifacts of this thesis were fine-tuned and evaluated using the Gemma 3-4B and Gemma 3-12B models. In addition to the research topic itself, the following AI tools were utilized in the process of creating the thesis. First, Grammarly was used for proofreading and grammar checking of thesis text drafts. Second, PyCharm IDE, including its integrated AI assistance tools along-side with Claude (Anthropic), were used during the implementation process to check codes, find errors, and for development support. In each instance, the AI generated suggestions were checked, validated and modified by the authors before being included. Manual annotation of grooming strategies was conducted by the two human annotators, without the use of AI, so that the ground-truth labels accurately reflect human decisions.

5.5 Conclusion

This study investigated on the research question “ How effectively can fine-tuned large language models identify specific grooming strategies in online chat conversations?”

The research question can be answered affirmatively. Fine-tuned large language models can identify specific grooming strategies more effectively than strategy-informed prompting alone. The fine-tuned model demonstrated improvements across multiple strategy classes and generalized across platforms, providing evidence that LLM-based strategy detection is a viable direction for advancing automated child protection.

The study made four principal contributions. First, it developed a five-strategy annotation framework grounded in established grooming theory [1, 2, 5] and demonstrated its reliability through inter-annotator agreement exceeding $\kappa = 0.80$ on PAN2012 and $\kappa = 0.77$ on LiveMe. This framework provides a reproducible instrument for future research on strategy-level grooming detection.

Second, it demonstrated that fine-tuning Gemma 3-4B on 120 manually annotated conversations improved strategy classification accuracy by 12.5 percentage points and enabled the detection of Sexual Normalization, a strategy that the baseline model failed to identify entirely. This finding confirms that human-annotated strategy labels provide a training signal that prompt-based definitions cannot replicate.

Third, it provided the first cross-platform evaluation of strategy-level grooming detection, showing that strategy representations learned from PAN2012 conversations transfer meaningfully to LiveMe conversations without additional training [4].

Fourth, the supplementary Gemma 3-12B comparison, where the 12B designation indicates a model with approximately 12 billion parameter compared to the 4 billion in the primary Gemma 3-4B variant revealed that model capacity influences baseline strategy detection capability, but that quantization may interfere with fine-tuning gains, informing future methodological choices.

However, the modest absolute performance levels and the limitations of small datasets indicate that this study represents a first step rather than a complete solution. Continued research with larger datasets, full-precision fine-tuning of larger models, and message-level annotation will be needed to realize the full potential of strategy-level grooming detection.

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Appendix A

conv_id	conversation_id_full	Primary_Strategy				Secondary_Strategy				Final_Arc Sexual Solicitation and Exploitationntence
		Annotator_1	Annotator_2	Agreement	Final_Annotation	Annotator_1	Annotator_2	Agreement	Final_Annotation	
1	be5f4efef03c787c548bc8580987b112	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator repeatedly escalates Sexual Solicitation and Exploitationxual conversation, pressures the minor toward Sexual Solicitation and Exploitationxual activity, and attempts to arrange an offline encounter in order to obtain Sexual Solicitation and Exploitationxual compliance through deSexual Solicitation and Exploitationnsitization, normalization, Sexual Solicitation and Exploitationcrecy, and coercive influence , which is intended to isolate the victim, reduce resistance, and
2	564c78a9844de46ee4773a9121df34cc	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator gradually introduces Sexual Solicitation and Exploitationxual topics and encourages the minor to view Sexual Solicitation and Exploitationxual activity as normal and desirable in order to reduce her inhibitions and prepare her for future Sexual Solicitation and Exploitationxual involvement , which is intended to increaSexual Solicitation and Exploitation her curiosity, lower resistance, and make her more
3	94d1eb0f36b4212486390f720170026f	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator blends romantic affection with sexualized fantasy prompts and encourages the minor to imagine intimate scenarios in order to gradually desensitize her to sexual content and normalize a sexualized relationship dynamic , which is intended to lower her boundaries, increase emotional dependence, and prepare her for future sexual
4	f9e287e20f56ea96cc668a1d2d45e563	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator blends intense romantic attachment with repeated sexual comments, explicit fantasies, and pressure to engage in sexual activity in order to desensitize the minor, normalize a sexualized relationship, and create emotional dependence , which is intended to lower her boundaries, increase compliance, and facilitate sexual exploitation
5	0b92a6b8f0471f599b8837409966bc3e	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator uses romantic affection, emotional pressure, and sexualized comments to build intimacy with the minor in order to normalize a sexualized relationship and increase emotional dependence , which is intended to lower her boundaries, encourage attachment, and prepare her for future sexual exploitation .
6	0d3d1be72fb34d3c2665fd6da201c16e	Sexual Normalization	Sexual Desensitization	Not Agree	Sexual Normalization	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Desensitization	The preadator introduces sexualized content, tests the minor's comfort level, and attempts to arrange an offline meeting in order to desensitize her to sexual topics and build trust for future escalation , which is intended to lower her boundaries and increase
7	520e0c52a8b4c229b5e989e816147b80	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator repeatedly introduces sexual topics, asks about the minor's sexual history, and encourages sexualized behavior and an offline meeting in order to desensitize her, normalize sexual involvement, and build compliance through emotional pressure and secrecy , which is intended to lower her boundaries and facilitate sexual
8	fd03cc88efa0fec3ab2c47b47c77374a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator abruptly introduces explicit sexual content and requests the minor to witness a sexual act in order to solicit sexual involvement despite her stated concerns about being too young , which is intended to pressure her into compliance and lower her
9	3abb1621cb55c3451e9020dce553bfd8	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator uses intense romantic language, emotional pressure, and promises of lifelong devotion in order to create dependency, normalize an adult–minor relationship, and prepare the minor for future intimacy , which is intended to lower her boundaries, increase compliance, and facilitate offline exploitation .
10	b687908b9eb40b9109b842ff21b98792	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Desensitization/Sexual Normalization	Not Agree	Control and Coercion	The preadator repeatedly introduces, escalates, and pressures the minor into agreeing to explicit sexual acts in order to normalize sexual behaviour, test boundaries, and secure compliance through manipulation and desensitization , which is intended to lower resistance, override autonomy, and facilitate sexual exploitation .
11	717885127ef7e9796bd6760e791730d4	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator repeatedly pressures the minor to meet for sexual activity, including attempts to persuade her to sneak out at night in order to normalize sexual involvement, override her hesitation, and secure compliance through emotional pressure , which is intended to lower her boundaries and facilitate offline sexual exploitation .
12	23fef4af1929764d65bea63b0dd25879	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator exposes the minor to sexual content, repeatedly requests sexual acts, and pressures her to perform them privately or on webcam in order to desensitize her, normalize sexual involvement, and secure compliance through emotional manipulation , which is intended to lower her boundaries and facilitate sexual exploitation both online
13	6b7572e126eb21facd2a39c738f885a1	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator uses sexualized compliments and requests for provocative images to normalize sexual attention toward the minor and test her boundaries , which is intended to increase her comfort with sexualized interaction and prepare for further exploitation .

14	64959c4f57a026fa3edca0c9b67202de	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Desensitization/Sexual Normalization	Not Agree	Sexual Normalization	The preadator uses sexual compliments, explicit descriptions, and plans for offline sexual contact to normalize sexual involvement with the minor, reduce her resistance, and frame the relationship as legitimate despite the age gap , which is intended to lower her boundaries and facilitate sexual exploitation .
15	afb26a3794e5aaeb68b407be7a6bb2c1	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator introduces sexualized comments toward the minor and frames them as affectionate and normal in order to reduce her sensitivity to sexual attention and maintain her engagement , which is intended to lower her boundaries and prepare for
16	0106d1a7ecccbd8a4510bed66121be1a	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator uses romantic and sexualized comments, requests for photos, and secrecy around communication to normalize inappropriate intimacy and maintain the minor's engagement despite parental oversight , which is intended to lower her boundaries and increase vulnerability to further exploitation .
17	ea4e82bbf029626622665239d15cda20	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator introduces and maintains sexual discussion, persistently proposes an offline sexual meeting, and pressures the minor to create opportunities for contact in order to normalize sexual interaction, reduce barriers, and gain compliance through manipulation and situational control , leading to increased willingness to meet, reduced
18	25e47a553ed6440e2fbbe759bdb7a	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator shares explicit content and pushes for sexual interaction to normalize and encourage sexual behavior , leading to increased acceptance and exploitation .
19	865a14a17fcceafb937149431de1d	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator introduced sexualized descriptions of the victim to shift the relationship into a sexual context , which is intended to lower her boundaries for future exploitation .
20	4a0e1ff2414267186b476255cfd12e2a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator repeatedly pressured the minor to undress and join sexual calls to push the interaction into a live sexual setting , which is intended to lower her resistance and enable direct exploitation .
21	4734beecd54f93cd472f750ca86005	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator used sexual comments and questions about nudity to shift the conversation into a sexual frame , which is intended to lower the minor's boundaries for
22	2136b96aad8a751428f86736ce1ee	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator pushed for private phone contact and secrecy to avoid detection while deepening the relationship , which is intended to create controlled access for future
23	62e282a0191e43bdc7feb7ee40c90	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushed for private in-person contact and personal details to secure unsupervised access , which is intended to enable a sexual meeting with reduced risk of
24	d5ee98309092b769f853217f71631	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator escalated sexual talk and probed for sexual details to normalize intimacy with a minor , which is intended to lower her boundaries for future sexual exploitation .
25	c29dc96758ca816f113d2b507f905	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator arranged a hidden in-person visit and used affectionate pressure to secure private, unsupervised access , which is intended to enable intimate contact
26	7f4df6c4fdd3b22e6becf9e0c585a47	Sexual Normalization	Sexual Desensitization	Not Agree	Sexual Desensitization	Sexual Desensitization	Sexual Normalization	Not Agree	Sexual Desensitization	The preadator used sexual talk and age-gap reassurance to build intimacy with the minor, aiming to reduce her resistance and enable future sexual contact.
27	68cb74a101e21b0fb03bde3adbc2f	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushed explicit sexual acts and pressured her to meet secretly, aiming to normalize exploitation and reduce her resistance to future sexual contact.
28	19491ab8958f7280d07bb572b1baa	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator used explicit sexual planning and pressured her to meet privately, aiming to normalize exploitation and make her more compliant during future sexual contact.
29	539fc3f7340da3b5e9b48a7e03320	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator framed sexual involvement as maturity and excitement to normalize intimacy with the minor, aiming to reduce her hesitation and prepare her for future
30	593ab3e70f995052949b21d0ee1db	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The preadator used emotional pressure and sexualized reassurance to normalize intimacy with the minor, aiming to keep her compliant and prepared for future sexual
31	238d89ae174c904a0292d23d2407	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator escalates explicit sexual demands to secure compliance, aiming to ensure the victim feels obligated to meet him and perform the acts he insists on.
32	6476472a0dcdd3fb855382e1e555d	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator uses sexualized questions and age-minimizing comments to build comfort with inappropriate intimacy, aiming to shift the minor toward future offline contact.
33	e1a832ea31d9b717748b1202b6fe4	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator arranges an in-person sexual encounter and reinforces it with explicit talk and affection, aiming to ensure the minor follows through with the planned meeting.
34	5c29d3611c1b72b98e6ab4ab49f47	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator blends romantic bonding with escalating sexual talk to lower resistance, aiming to deepen emotional dependence and normalize increasingly explicit
35	90250ba82cd50df97a7e53e3b8f0ed	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Desensitization	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Desensitization	The preadator pressures the minor into planning a sexual meeting and explicit acts, aiming to ensure secrecy and compliance when the opportunity arises.
36	9b960a4e97957941ce768c6f856e9	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator introduces sexual interest immediately to lower boundaries, aiming to normalize attraction to minors and prepare for further escalation in later interactions.
37	b16bf14b126f5d51e369053f2211d9	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator uses affectionate language and sexualized compliments to lower boundaries, aiming to normalize ongoing image sharing and deepen the minor's

38	aa98783f74a399c4e1f1415893b0f1	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator uses affectionate flirting that shifts into sexualized kissing to lower boundaries, aiming to normalize intimate behavior and deepen the minor's emotional
39	03d25d2cf3768f2d5745f49a520638	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator builds secrecy and emotional dependence to arrange an in-person meeting, aiming to ensure the minor's compliance while avoiding detection by adults.
40	ab30ad1266a8a6070a1bb3e134a5	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Desensitization	The preadator escalates explicit sexual demands while coordinating a secret meeting, aiming to secure compliance and create opportunities for recording and exploiting the
41	0b5869b9bbcc3399676815ee6619	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Desensitization	The preadator uses explicit sexual talk, live sexual displays, and romantic pressure to secure compliance, aiming to ensure the minor participates in planned sexual activity.
42	320f7cd503e78396dcbcaa8cec03c8	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator uses secrecy, emotional pressure, and explicit sexual planning to secure compliance, aiming to ensure the minor participates in a multi-day sexual encounter.
43	de65c8732fb505e2763885c3ef6a11	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses explicit sexual questioning and secrecy to escalate intimacy, aiming to secure compliance for an in-person sexual encounter while minimizing the risk of
44	352acde13b5267574985b2d67597	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses explicit sexual coaching and secrecy to escalate intimacy, aiming to guide the minor into sexual acts and prepare for further private contact.
45	7ea9e30d5b998147d269bff5756b8	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator escalates explicit sexual dialogue and pushes for physical contact to increase compliance, aiming to create opportunities for direct sexual access and
46	396268354ad11439683772b168218	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator intensifies sexual dialogue and pushes for explicit interaction to build compliance, aiming to transition the online dynamic into direct sexual access and
47	0488b1c439644f1722839ff39b899e	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator pushes explicit sexual expectations and arranges an in-person encounter to secure compliance, aiming to isolate the minor and facilitate direct sexual
48	ead5e7245c28c136fe0c7d52f14ab	Isolation and Secracy	Sexual Solicitation and Exploitation	Not Agree	Isolation and Secracy	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator probes for a discreet meeting location to initiate in-person contact, aiming to reduce oversight and create an opportunity for potential sexual exploitation.
49	91ec8c9b4ade6231d3b5ae7a578c	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator overwhelms the minor with explicit sexual content and requests to increase compliance, aiming to normalize sexual interaction and prepare for potential
50	4aaafb2dcf62a9eea8d6ac69aca5c6	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator escalates sexual dialogue and pushes for an in-person encounter to secure compliance, aiming to isolate the minor and facilitate direct sexual exploitation.
51	676f446db9bb0a7f46975d048e5e9a	Sexual Normalization	Sexual Solicitation and Exploitation	Not Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator uses explicit sexual instruction and promises of "teaching" to shape compliance, aiming to prepare the minor for sexual access and reduce resistance during
52	d895be43ffab826aed8b1d909d5fdd	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator directs explicit sexual planning and pressures the minor to appear older to avoid detection, aiming to isolate her and enable direct sexual access.
53	99d4d9d1f8cb537a8201d1cb775ec	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator blends emotional intimacy with sexual curiosity to increase compliance, aiming to normalize sexual interaction and create opportunities for private contact and
54	fbfda57dae3045510ed9fa8e3b758e	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator encourages secrecy and sexual experimentation while arranging an in-person meeting, aiming to reduce oversight and create an opportunity for direct sexual
55	cce60f9689a2210b9a5cc0a8687c49b1	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses romantic framing and future-oriented promises to normalize sexual expectations, aiming to secure emotional dependence and facilitate eventual in-person sexual access.
56	c4e1ef1b0e565363a1f8580474314d	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The preadator uses coercive sexual demands and emotional pressure to secure compliance, aiming to isolate the minor and ensure she submits to planned in-person
57	72f8843c2e088a6d4954ed48653ee	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses escalating sexual comments to lower the minor's resistance, aiming to normalize explicit interaction and prepare her for future sexual compliance.
58	bb362da5b74e982b230302a5f0370	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator aggressively pushes for sexual acts and an in-person encounter to secure compliance, aiming to normalize sexual access and exploit the minor during a planned
59	dc5569be77ec4ea23cef9e7ceb1e8	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator uses emotional closeness to push for sexual agreement, aiming to normalize sexual involvement and prepare the minor for future sexual exploitation.
60	41a3b638e68781017b285d6259a2f	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator reframes the relationship as romantic and appropriate while encouraging secrecy, aiming to normalize emotional dependence and prepare for future private
61	ab42a85c6dd371e3bf98475d5642a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator escalates sexual requests and plans an in-person encounter, using explicit detail to desensitize the minor and prepare her for direct sexual exploitation.
62	59254d59cbe01d1b1f6bceea8845b	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses manipulation and emotional pressure to control the minor, pushing for secrecy and arranging an in-person meeting intended for sexual exploitation.
63	5fac4012fbf496ce9d462c20d0c9f9d	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator pressures the minor to meet privately while requesting sexualized clothing, aiming to isolate her and set up an in-person encounter for sexual exploitation.
64	1a53289d3329b6d146326c1a32c3	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses explicit sexual questioning and graphic descriptions to lower the minor's resistance, aiming to prepare her for future sexual acts and exploitation.
65	35efcc84a3657656badeab2a7b453	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator intensifies explicit questioning and plans sexual activity to condition the child into compliance, aiming to secure willingness for escalating sexual exploitation.

66	76ddf028c88703f1db859b760fda49	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Desensitization	Not Agree	Control and Coercion	The preadator persistently pressures the child toward sexual communication and an in-person encounter to secure compliance, aiming to establish exclusive access for
67	a97e989ff53c7256b77bad22993c4	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator aggressively pushes explicit sexual acts and fantasies to secure compliance, using normalization to erode boundaries and increase the child's
68	674f281bfbfcc7ca86a514e826f243c	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes explicit sexual acts and a planned visit to secure physical access, using escalating sexual dialogue to reduce resistance and increase compliance for
69	ad2830164c55b0ecfa3d06669ad15	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator gathers personal details and tests boundaries to assess privacy, using mild pressure about legal concerns to shape the child's dependence and future
70	5b5041b9acf01636121f5226885f45	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator gradually introduces sexual intimacy to lower boundaries, then pushes for an in-person meeting, aiming to secure private access for escalating exploitation.
71	456e1f8cfc67f1cae1261055dfe92c5	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator intensifies sexual talk and explicit touching fantasies to lower resistance, then solicits sexual participation, aiming to secure the child's compliance for escalating
72	b0d9cfcf8abd2cddb9399ad4d20b2	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator introduces romantic physical contact to lower boundaries, then encourages the child to agree to making out, aiming to increase comfort with escalating
73	b32eb7fec965d83d66f1bb9787543a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes explicit sexual acts and plans for future contact, using escalating sexual talk to reduce boundaries and increase the child's willingness for exploitation.
74	ae3373d2f034079e6d38001647222	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator directs explicit sexual acts and webcam behavior to gain sexual access, using escalating sexual talk to reduce resistance and secure ongoing compliance for
75	7b861f097a29825d6b3d45d963b22f	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator introduces flirtatious physical intimacy to lower boundaries, then encourages kissing and closeness, aiming to increase the child's comfort with escalating
76	9fdca5f57609c78e3704af9913611f	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes for an in-person sexual encounter and directs the child to prepare sexually, using escalating sexual talk to reduce boundaries and secure compliance.
77	7716dce5ec329662f4cfd86bf1e869	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	The preadator uses sexual teasing to lower boundaries, then arranges a private meeting timed around parental absence, aiming to create an opportunity for sexual access.
78	01c87a646ae83beb5aba08de0eed	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator encourages secrecy and probes for personal details to isolate the child, using protective framing to influence decisions and increase vulnerability to future
79	9799dd7ab882fe1293ec1e7afd9dba	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes explicit sexual acts and a planned meetup, using escalating sexual questions to lower boundaries and secure the child's willingness for exploitation.
80	ec88307dbb7bb6f952ba100842849	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes explicit sexual acts and a planned meetup, using escalating sexual talk to lower the child's boundaries and secure compliance for in-person exploitation.
81	c3119e0b4d26c337f2b147e0b3f09	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pushes extreme sexual acts, including bestiality, and arranges an in-person encounter, using escalating explicit talk to erode boundaries and secure
82	13d0203b773fc4bd99d96bb3f97431	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator drives explicit sexual acts and future contact, using escalating sexual instructions to erode boundaries and secure the child's compliance for ongoing
83	4aca749f11771c8b069710577948e	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator sexualizes affection through teasing and spanking, normalizing sexual contact as playful and loving to reduce boundaries and prepare the child for future
84	fc7e2594b1013975337df5cab08830	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	The preadator coerces explicit sexual acts, including incest, using escalating instructions and pressure to erode boundaries and secure the child's compliance for
85	a2adb2a63cc9bbd2d45f477a0ec5c	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The preadator pressures the child toward intercourse and future sexual contact, using escalating romantic-sexual talk to lower boundaries and normalize sexual involvement.
86	a0abe3985d3a37424eef24dbcb241	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The preadator acknowledges the victim's young age while minimizing concern and using emotional reassurance to preserve rapport, intending to maintain engagement despite
87	e417ee1a6cfc39f73153657ab2de6	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender intensifies sexual dialogue and arranges an in-person encounter to secure compliance, aiming to transition the relationship into a sexual encounter for personal
88	9ece41d168effa734b002ab3eed7c	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender normalizes the age gap and encourages private contact to create opportunities for unsupervised access, aiming to shift the relationship toward in-person
89	60691a7fd740b10bd301568f4fc9de	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Normalization	Not Agree	Sexual Desensitization	The offender escalates explicit sexual dialogue and pressures the minor toward physical compliance to secure sexual access, aiming to normalize exploitation and ensure future
90	98f2447e29ee9b9db68e8729fbfbf60	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates sexual questioning and arranges unsupervised access to secure compliance, aiming to transition the minor from online sexual engagement to in-person
91	e05454e8ea60273e0e68cc5ff49b7059	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender blends romantic affection with escalating sexual comments to normalize intimacy, aiming to deepen emotional dependence and reduce the minor's resistance to future sexual advances.
92	3878b7b81a0efec756fefa39e24dd8	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender rapidly escalates explicit sexual questioning to normalize graphic sexual interaction, aiming to exploit the minor's disclosures and maintain ongoing sexual

93	f425d9d4b903780d4f7c7a1747d10c3f	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender uses intimate kissing prompts and private communication requests to sexualize interaction and secure unsupervised access, aiming to deepen vulnerability and enable future exploitation.
94	497723ecce35759c070c58d13a5c07aa	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender uses intimate kissing prompts and private communication requests to sexualize interaction and secure unsupervised access, aiming to deepen vulnerability and enable future exploitation.
95	0c671d77f89c29dc672b1688a4835	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	The offender escalates graphic sexual dialogue and pushes for explicit photos to normalize exploitation, aiming to secure compliance and prepare for future in-person
96	6b131fb12653e7d956d89699be2f00	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	None	Sexual Normalization	Not Agree	Sexual Normalization	The offender introduces romantic physical intimacy to normalize sexual closeness, aiming to build comfort with escalating affection and prepare the minor for future sexual
97	96e20f1c261a48a14786d67a552e0	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Desensitization	Not Agree	Control and Coercion	The offender escalates explicit sexual demands and pressures the minor to perform sexual acts, aiming to normalize exploitation and maintain ongoing sexual compliance
98	9b86056fd55117a1838a8d1916a52	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender arranges a private multi-day hotel stay to secure unsupervised access, aiming to normalize the encounter and prepare the minor for in-person exploitation.
99	84ceb10d78594dbdfe9358967d5d5	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender downplays the age gap and uses playful affection to normalize closeness, seeking visual access that can later support more intimate, boundary-crossing
100	9d1bf5b626861562e2b307d367fff1e	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender uses emotional pressure and romantic framing to maintain closeness, aiming to secure compliance and deepen dependence despite the minor's hesitation
101	e400cc1a2a30f9b350ae73c0f1601f	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender uses flirtation and minimizes the age gap to normalize inappropriate closeness, aiming to build comfort and prepare the minor for future boundary-crossing
102	2c1db7060ff800950f27a23f8f3711f	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates sexual dialogue and arranges secret offline contact to normalize sexual involvement with a child, aiming to secure compliance for explicit physical acts.
103	69715c391bebe32a07dde0f3e049d	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender cultivates emotional intimacy and reframes sexual involvement as love to erode boundaries, aiming to secure the child's willingness for future sexual contact.
104	0b5c826cfd028f766c1586c35b70f1	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender pressures the child to reassure him and agree to meet, using emotional manipulation to reduce resistance and increase compliance for an in-person encounter.
105	e3e2daa1b3db76a3bd2c87102915	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender sexualizes casual conversation and requests explicit images to build compliance, aiming to normalize sexual interaction and prepare the child for future
106	d8e9382a292f3fa6e0b3b9872fb2ee	Sexual Desensitization	Sexual Normalization	Not Agree	Sexual Desensitization	Sexual Normalization	Sexual Desensitization	Not Agree	Sexual Desensitization	The offender romanticizes and sexualizes the minor to redefine the relationship as acceptable, aiming to lower boundaries and increase emotional vulnerability for future
107	7a99f624c6a25c57107ba24aba73c	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender repeatedly describes sexualized nudity and exposure to normalize explicit behavior, aiming to lower boundaries and increase the minor's comfort with sexual
108	83b7266ccd5477930d22b9900cdc	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender escalates explicit sexual dialogue and arranges an in-person encounter to secure compliance, aiming to transition the minor from online grooming to physical
109	7a4ef3fc59a4edbdb8b699d9c0b6c	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender blends romantic affection with explicit fantasies to normalize sexual involvement, aiming to deepen emotional dependence and prepare the minor for an
110	8b1ab967808bdb0ef86cd7a9babad	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates explicit sexual fantasies and romantic framing to normalize sexual involvement, aiming to secure the minor's agreement to meet for physical sexual
111	ac9d9f221e87f40257af21d68303ce	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender uses coercive sexual demands, emotional manipulation, and conditional rewards to normalize exploitation, aiming to secure the minor's obedience and prepare
112	776ac255556ca2d6ccfb15d48b93e	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender pressures the minor for secrecy and sexual access, aiming to normalize intimate contact and secure a private meeting for future exploitation.
113	e74118c810139926864ea9a25b28	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender uses sexual dominance, emotional ownership, and escalating explicit demands to normalize exploitation, aiming to secure total compliance and long-term
114	11be31930917f18ef6d41d0bc7bff89	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender pressures the minor for secrecy and emotional compliance while introducing sexual expectations, aiming to normalize exploitation and secure control
115	ab3aefbbdbdfe490804c0339c63e5	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Isolation and Secrecy	Not Agree	Sexual Normalization	The offender introduces mild sexual scenarios and encourages secrecy around communication, aiming to gradually lower boundaries and prepare the minor for future
116	b074c8739d57ce09e532812eb249	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender introduces sexual frustration and cam-related comments to gradually lower boundaries, aiming to normalize sexual discussion and prepare the minor for future
117	ca2fbadd78607459cef6b1a66a1fe267	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender intensifies sexualized compliments and body-focused questions to lower boundaries and reframe sexual attention as normal, aiming to increase the minor's comfort with escalating sexual requests.
118	646c4e61b0bd1adf3e689c2465ef3	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates explicit fantasies and frames sexual behavior as normal to lower resistance and exploit her inexperience, aiming to increase compliance with sexualized
119	40e979e5610771 4544bd72f245220	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender uses escalating sexual imagery and propositions to lower boundaries and normalize intimacy, aiming to draw the minor into increasingly explicit and secretive

120	22a01ba30004277eccbf3bb5c72c1	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	“The offender intensifies sexual talk and proposes an in-person encounter to normalize intimacy and escalate compliance, aiming to transition the minor into direct sexual
121	49931b17c5cc78f7c70d7503f1c075	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses graphic sexual disclosures and exposure to lower boundaries and provoke engagement, aiming to normalize explicit behavior and draw the minor into
122	94c71d9e905c390d310f3f315f9c7b19	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation/Sexual	Sexual Normalization	Not Agree	Sexual Normalization	The offender pushes for a secret in-person meeting and minimizes the age gap to secure compliance, aiming to isolate the minor and normalize prohibited contact.
123	66150f5e469e7b841e245b0dfdeefa	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	None	Sexual Normalization	Not Agree	Sexual Normalization	The offender instructs the minor to erase chat records and hide communication, aiming to eliminate oversight and maintain a secret relationship that bypasses adult detection.
124	0eb971370f1fed2b57fe5950b2c7f6	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The offender arranges a secret in-person meeting and demands sexualized compliance, aiming to isolate the minor and ensure she follows his escalating sexual expectations.
125	34e1f8ebd0afe0c13fc93c1bdcdc72	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender introduces sexual topics and probes comfort with intimate behavior to lower boundaries, aiming to shift the minor toward accepting increasingly explicit
126	816e563be454ccc9c25207236196	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Solicitation and Exploitation	Not Agree	Control and Coercion	The offender escalates graphic sexual demands and pressures the minor to meet in person, aiming to normalize explicit acts and secure compliance with his sexual
127	c94ff7ff76012617c574ff8bb7a97cc	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates romantic talk into explicit sexual demands to lower boundaries and push toward an in-person encounter, aiming to secure the minor's compliance with
128	1c548f7b9f37de8ef09d10f3c3db664a	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender uses romanticized future promises and planned sexual contact to normalize an illicit relationship, aiming to secure the minor's emotional dependence and compliance with meeting in person.
129	5f0cbc1327625752d60d15ba5c432	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates explicit sexual acts and pressures the minor to create sexual recordings, aiming to normalize exploitation and secure compliance with increasingly
130	db8103620bc75105bcb1f898d0a57	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	“The offender arranges a secret meeting and pressures the minor to trust him, aiming to isolate the child and prepare for sexual exploitation during the planned encounter.
131	caef747034ddcfcb7a8e2e0cbcd0fc	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates sexualized questions and explicit arousal descriptions to lower boundaries and shape the minor's reactions, aiming to increase comfort with sexualized
132	b4c3a2538ca6083f84420a95a591eab7	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender escalates explicit sexual dialogue and pressures participation in planned sexual acts to reduce resistance, enabling offline exploitation and increasing compliance during future encounters.
133	ca8304c6422a5cd6e85e106413e3	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender normalizes a large age gap and subtly sexualizes the interaction to reduce boundaries, aiming to increase the child's comfort with future inappropriate advances.
134	0acdf02f3b580b79bfb8997ed6eb69	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender rapidly escalates explicit sexual demands and pushes for covert offline contact to override boundaries, aiming to secure physical access and compliance.
135	9b47ba64d2eaec1e8f4b0ccde4f95	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender pressures the child into committing to a sexual meeting and reinforces explicit expectations to reduce resistance, aiming to secure compliance and ensure
136	8c3427b592030839a3334777e783	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates explicit sexual dialogue and directs the child toward private sexual compliance to lower boundaries, aiming to increase vulnerability and future
137	1003058f2eca1b53af9551128dd4a4	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender intensifies explicit sexual dialogue and reframes sexual activity as desirable and inevitable to reduce resistance, aiming to secure agreement for an offline sexual
138	2a04ae4fd43ef77041ed8fa541a5dc	Control and Coercion	Control and Coercion	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation	Sexual Normalization	Not Agree	Sexual Solicitation and Exploitation	The offender pressures the child to arrange an unsupervised overnight visit and frames it as romantic closeness, aiming to reduce resistance and secure private access.
139	1c7f99d37b2e1ad19e9844affccbf79	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender encourages secrecy and manipulates the child into hiding his age to enable unsupervised contact, aiming to secure private access and reduce adult intervention.
140	b4470f0c3a19766054b8395463356	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender uses explicit sexual fantasies and requests for personal images to lower boundaries, aiming to increase the child's comfort with sexualized interaction and future
141	9ba9cd983854cbfeac5b0fc514cb90	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates explicit sexual requests, normalizes sexual behavior, and pressures the minor with romantic framing to condition compliance and enable future
142	32cabf2e39899add8a27d42dac229	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender orchestrates a secret in-person sexual meeting, escalating explicit questions and secrecy to secure compliance and create conditions for direct sexual
143	404c9a33d5acd75de1471dcc267a	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender normalizes sexual involvement with a minor while positioning himself as uniquely accepting, aiming to lower boundaries and prepare for future sexual
144	0c1cfc7c43c94d0c2548438798ceff	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender pushes secrecy, hidden photos, and covert meetings to isolate the minor and escalate sexual access while reducing the chance of outside interference.
145	5a038afb69efe3f249615652f3c6cba	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates sexual talk, requests images, and performs explicit acts on camera to condition compliance and deepen exploitation while urging secrecy to avoid
146	57dcd727c4e46f13937a94a820814	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses explicit sexual talk, grooming instructions, and pressure for nude photos to lower boundaries and exploit the minor while encouraging secrecy to avoid

147	ef5b2b02d038fc3e8706faeddccb64	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The offender uses secrecy, emotional pressure, and sexualized questioning to deepen dependence and arrange a covert meeting intended to enable exploitation while avoiding
148	67d9eb94501991259e77bc45d15b	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender reframes sexual activity as meaningful and honorable, using emotional pressure and explicit fantasies to lower resistance and secure future sexual compliance.
149	1ce8fe1c70214391be6eb725cf8d49	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender uses explicit sexual promises, emotional bonding, and secrecy to prepare the minor for in-person sexual exploitation while reinforcing dependence and
150	8cf56dee21c29ded112679ca1cf8f8	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses emotional pressure, explicit sexual promises, and normalization of unprotected sex to secure compliance and prepare the minor for in-person sexual
151	cc5fba01f752fae4846aed8f26731b7	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Solicitation and Exploitation	Not Agree	Control and Coercion	The offender intensifies explicit sexual dialogue and pushes for in-person sexual activity to shape the child's expectations, enabling easier exploitation during a planned offline
152	78cd7282266f556846320b8a2637a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The offender pressures the child for secrecy and arranges an unsupervised visit to maintain control, enabling a private environment that facilitates planned exploitation.
153	b6469e53c067e61c50bb2db07bcf1	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates explicit sexual roleplay and pressures the child to describe intimate acts to condition compliance, enabling smoother transition toward planned
154	89149422447aa914e3294d222242	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender proposes an in-person visit and sexualized shower scenario to condition compliance, creating a private setting intended to facilitate direct physical exploitation.
155	29cb24377cee9531055337d0de6a	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender probes for parental absences and arranges private access to the child to create unsupervised opportunities intended to enable future exploitation during
156	21ef792cffe154b94448115ffd91714	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender escalates extreme incest-themed sexual dialogue to reshape the child's boundaries, aiming to normalize exploitation and secure unquestioning compliance for
157	c65067fcc33771649fc5b53a4a6d1c	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender pushes sexual questions and webcam requests to lower the child's boundaries, aiming to normalize explicit behavior and create opportunities for escalating
158	0c34ab23c7abb91c8728c078cb1b	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender introduces romanticized sexual fantasies to lower the child's boundaries, aiming to normalize intimacy and create emotional dependence that facilitates future
159	6eab795c5f6a9d822d25a2b153736	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender engineers secrecy and arranges a covert meeting while escalating sexual questioning to assess vulnerability, aiming to secure private access for direct
160	8da40d7803e151a4c5566f9203bf4	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The offender blends affection with sexualized requests to lower boundaries, aiming to normalize intimate behavior and secure secret communication that enables future
161	a4c57c2dd047a11f2a185f70d44bb	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses intense romantic bonding and future-focused sexual framing to reshape the child's expectations, aiming to secure emotional dependence that enables
162	731da0287a60fea99bfff0af9b58873	Sexual Solicitation and Exploitation	Sexual Desensitization	Not Agree		Control and Coercion	Sexual Solicitation and Exploitation	Not Agree	Control and Coercion	The offender escalates explicit sexual dialogue and orchestrates a threesome scenario to erode boundaries, aiming to secure secret compliance and facilitate direct sexual
163	952c719c2ff7d73eed4455e7decf52	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender plans multi-day unsupervised visits and embeds sexual expectations into affectionate talk to erode boundaries, enabling private access intended for escalating
164	f9b668563394cef891044afdd2d402	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender blends romantic affection with sexualized compliments to lower boundaries, aiming to normalize intimacy and encourage private contact that enables
165	008862aa127e7d996e75cda14101	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender arranges secret in-person access and embeds sexualized affection into routine conversation to lower boundaries, aiming to secure private contact for future
166	dfc09bf91312711e5656c2ba6433f7	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender intensifies explicit sexual demands to condition compliance, aiming to secure escalating sexual access that reinforces dependence and reduces the victim's
167	aa8750305414c2a60076e518ed9b	Isolation and Secracy	Isolation and Secracy	Agree	Isolation and Secracy	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender pressures the child into secrecy and emotional dependence to secure an in-person meeting, aiming to reduce outside interference and increase vulnerability.
168	7abc38b5511fe59bc6b589904bd35	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates sexual talk and pressures the child toward physical contact to secure an in-person encounter, aiming to increase compliance and reduce resistance.
169	0518fe715b9941c5da38c8123f4dbe	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The offender escalates explicit sexual planning and coordinates secrecy to secure an in-person encounter, aiming to increase vulnerability and ensure compliance during the
170	1e0fffe493a8c907c82bbdc62c022e	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses explicit images and possessive romantic framing to push the child toward sexual compliance, aiming to secure further exposure and future physical
171	8ff0f467dde1f1eca174920da4016b	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses explicit sexual instruction and pressure to push the child toward sexual experimentation, aiming to prepare her for future physical access and reduce
172	ef76a4ed8624740d1e53193c3f9cc3	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses escalating sexual pressure, emotional manipulation, and possessive framing to shape the child's compliance, aiming to secure total control and future sexual
173	020f9a104c1498cf7ed3664c7f45cc	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender blends sexualized affection with secrecy and emotional closeness to maintain influence, aiming to normalize sexual interaction and preserve ongoing access.
174	727f3118c5536f8f35c97cca98c584	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender orchestrates a secret in-person meeting and pressures the child toward sexual activity, aiming to ensure compliance while minimizing the risk of discovery.

175	0ba01743d7d9a7daf9b9004d4f13f1	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Sexual Desensitization	Not Agree	Isolation and Secrecy	The offender intensifies explicit sexual questioning and encourages secret masturbation to condition compliance, aiming to prepare the child for future sexual access and reduce
176	92c2d716399fd44ffd3547ac8d4cee	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Desensitization	Not Agree	Sexual Desensitization	The offender escalates explicit sexual plans and pressures compliance to normalize extreme acts, aiming to secure an offline encounter that increases vulnerability and
177	8dbbb83731ebbdaee9623ac0b797	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender introduces romanticized sexual imagery to build comfort with intimacy, aiming to deepen emotional dependence and prepare the child for more explicit future
178	f4006dba4a4d31705c826f14e020fd36	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender uses escalating sexual violence, coercive fantasies, and abduction scenarios to erode boundaries and secure total compliance, aiming to establish complete control and ongoing exploitation.
179	ce9040bd56371a59ca0cb5b745cb	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender uses romantic framing and logistical planning to normalize an illicit meeting, aiming to create private access that increases dependence and reduces
180	762e27735ea44aebafe2ce8e15849	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates explicit sexual expectations and compares the child to prior minors to normalize compliance, aiming to secure unsupervised access and ongoing
181	6e76a41bc50b73c6d1d5a686dfae8	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Control and Coercion	Agree	Control and Coercion	The offender uses romantic framing and explicit sexual promises to normalize exploitation, aiming to secure a secret in-person encounter that increases vulnerability
182	85fe19e13903f7f6135907542f63dcd	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender blends romantic reassurance with explicit sexual planning to normalize exploitation, aiming to secure private access and deepen the child's emotional and
183	87b4adf826335bdfb7cc699f11c312	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender escalates explicit sexual expectations and pressures compliance through reassurance and secrecy, aiming to secure private access and increase the child's
184	d8ce51ff13bf9fbf667eb4b1d32498	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses escalating sexual descriptions and planned physical contact to normalize exploitation, aiming to secure private access and deepen the child's
185	2aac563c0ebde59b5e152c3f1a276	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender introduces mild sexual nudity and teasing to test boundaries, aiming to increase comfort with sexualized interaction and prepare for potential escalation.
186	2f2d891a4233354155f63e9b42901f	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and Coercion	Sexual Desensitization	Not Agree	Control and Coercion	The offender uses romantic reassurance and explicit sexual instruction to normalize exploitation, aiming to secure private access and increase the child's emotional and
187	23ac030176a90d034a696e29f2b7e	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses emotional closeness and escalating sexual fantasies to normalize exploitation, aiming to increase the child's readiness and anticipation for future sexual
188	9344a090cb15bfcccf737411ab6f1d	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates explicit sexual questioning and instruction to normalize exploitation, aiming to increase the child's compliance and prepare for future sexual
189	bdcb394cafe72d726e2ce7978c4efe	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender mixes explicit desire with legal justifications to normalize sexual interest, aiming to maintain emotional influence while avoiding direct commitment to physical
190	f81c4b28061bb24d0c0b9df26938a1	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates explicit sexual interaction and encourages the child to reciprocate, aiming to deepen compliance and prepare for future offline sexual access.
191	99f68760c5d83b43339ee9dd72118	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses explicit sexual fantasies and coordinated secrecy to lower resistance and secure agreement for offline sexual contact, aiming to ensure access and
192	b79a4cf30dee147f6a01ff855b1b836	Sexual Solicitation and Exploitation	Control and Coercion	Not Agree	Control and Coercion	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	The offender pressures the child to arrange an unsupervised meeting and dictates sexualized preparation, aiming to ensure privacy and facilitate immediate offline sexual
193	8b756433892714662fa9b7699b719	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender uses escalating sexual talk and plans for unsupervised contact to lower boundaries and secure compliance, aiming to create private opportunities for sexual
194	772fc5f5c93cc3c7994c902643abb	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender pressures the child to share photos and arrange a covert meeting, using sexual comments to lower resistance and secure compliance for offline sexual access.
195	d7c7ad048eb4f4b6ff80cdf41e33b4a	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The offender pushes explicit obedience-based sexual behavior to condition compliance, enabling escalating control and ensuring the child accepts his authority for future sexual
196	f34a53ffe68add795bf2cd8ecd2ea3f	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates sexual talk, secrecy, and photo requests to condition compliance and secure a covert sexual meeting, enabling further exploitation without outside
197	9d0f84ac5cdb2e1ce940cc13702af9	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	The offender uses sexual jokes, photo pressure, and plans for an in-person meeting to gradually condition compliance and create opportunities for future exploitation
198	3e2d5e4478272da6ecf2fbe958cd0	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender escalates sexual talk, lingerie planning, and photo requests to condition compliance and create a sexualized dynamic intended to enable future exploitation
199	cb76d063d0ba52fa3f5eeeddb68dfb	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	None	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	The offender downplays the minor's age and normalizes participation in a sexual chat space to test boundaries and create conditions for future sexual engagement
200	b418f0785c94df96bbaf55c2bdd627	Control and Coercion	Control and Coercion	Agree	Control and Coercion	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The offender uses emotional dependence and guilt to draw the minor into a private, exclusive bond intended to increase influence and reduce outside support

Appendix B

Results

Rater Agreement

Cohen's kappa

			95% CI	
Ratings	kappa	SE	Lower	Upper
Average kappa	0.874			
Annotator 1 - Annotator 2	0.874	0.029	0.817	0.931

Note. 200 subjects/items and 2 raters/measurements.
Confidence intervals are asymptotic.

Fleiss' kappa

			95% CI	
Ratings	Fleiss' kappa	SE	Lower	Upper
Overall	0.874	0.044	0.789	0.959
1	0.856	0.071	0.717	0.995
2	0.856	0.071	0.717	0.995
3	0.904	0.071	0.765	1.043
4	0.869	0.071	0.730	1.008
5	0.922	0.071	0.783	1.061

Note. 200 subjects/items and 2 raters/measurements.
Confidence intervals are asymptotic.

Krippendorff's alpha

			95% CI	
Method	Krippendorff's alpha	SE	Lower	Upper
Nominal	0.874	0.028	0.817	0.926

Note. 200 subjects/items and 2 raters/measurements.

Results

Rater Agreement

Cohen's kappa

			95% CI	
Ratings	kappa	SE	Lower	Upper
Average kappa	0.834			
Annotator 1 - Annotator 2	0.834	0.032	0.772	0.896

Note. 200 subjects/items and 2 raters/measurements. Based on pairwise complete cases. Confidence intervals are asymptotic.

Fleiss' kappa

			95% CI	
Ratings	Fleiss' kappa	SE	Lower	Upper
Overall	0.834	0.040	0.755	0.913
1	0.803	0.071	0.663	0.943
2	0.914	0.071	0.774	1.054
4	0.781	0.071	0.641	0.921
3	0.923	0.071	0.783	1.063
5	0.742	0.071	0.602	0.882

Note. 197 subjects/items and 2 raters/measurements. Based on listwise complete cases. Confidence intervals are asymptotic.

Krippendorff's alpha

			95% CI	
Method	Krippendorff's alpha	SE	Lower	Upper
Nominal	0.835	0.032	0.766	0.895

Note. 200 subjects/items and 2 raters/measurements. Based on pairwise complete cases.

Appendix C

conv_id	Primary_Strategy				Secondary_Strategy				Final_Arc _Sentence
	Annotator_1	Annotator_2	Agreement	Final_Annotation	Annotator_1	Annotator_2	Agreement	Final_Annotation	
201	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator is pressuring the victim to share explicit videos eventhough victim refuses and try to manipulate her by form of gift giving.
202	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator is making direct request for a sexual role play via online and continuously pressuring the victim to perform emotionally also aggressively.
203	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Preadator is making direct request for explicit videos and pressure victim to engage with him and also try to normalize the conversation via requesting the victim to be a girlfriend and giftgiving.
204	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator gradually escalates sexual contents to lower the victim's boundries in form of admiration of the victim's body features.
205	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator is making a friendly conversation with the victim.
206	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Preadator is gradually escallating with sexual content as compliments and sharing experiences with the victim.
207	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Preadator is gradually building trust with victim to separate and manipulate the victim to listen to him and starts to escalate with sexual content.
208	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Desensitization	Not Agree	Sexual Desensitization	Preadator is manipulating the victim to engage on a sexual roleplay via online and complimenting on her performance and offering rewards.
209	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Normalization	Sexual Normalization	Agree	Sexual Normalization	Preadator gradually escalates sexual contents to lower the victim's boundries and share experience and normalize the relationship.
210	Sexual Normalization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	Control and coercion	Sexual Desensitization	Not Agree	Control and coercion	Preadator is trying to connect with the victim emotionally, and try to normalize the friendship and behaviors.
211	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	None	None	Agree	General	Preadator is having friendly conversation with the victim, cannot decide any strategy directly.
212	Sexual Desensitization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	Sexual Normalization	Sexual Desensitization	Not Agree	Sexual Desensitization	Preadator gradually escalates sexual content as a form of truth or dare game to lower the victim's boundries.
213	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator is gradually escallating with sexual content and try to normalize the conversation.
214	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Preadator is gradually escallating with sexual content and try to lower the boundries.
215	Control and coercion	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Control and coercion	Not Agree	Control and coercion	Preadator is using emotional mannipulation, aggressiveness and threats at the same conversation to pressure the victim to agree on explicit content via online platform.
216	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Control and coercion	Control and coercion	Agree	Control and coercion	Preadator is trying to stablish secrecy on their conversation, trying to manipulate victim into being secretive.
217	Control and coercion	Isolation and Secrecy	Not Agree	Control and coercion	Isolation and Secrecy	None	Not Agree	None	Preadator is requesting victim into agreeing with a phone call.
218	Control and coercion	Control and coercion	Agree	Control and coercion	Sexual Normalization	Isolation and Secrecy	Not Agree	Isolation and Secrecy	Preadator is showing possessivenss and angry with the victim for not being agreeing with his terms.
219	Control and coercion	Isolation and Secrecy	Not Agree	Isolation and Secrecy	Sexual Desensitization	None	Not Agree	Sexual Desensitization	Preadator is inviting the victim to connect via other social media platform for a more private conversational opportunities.
220	Sexual Desensitization	Isolation and Secrecy	Not Agree	Isolation and Secrecy	Sexual Normalization	Sexual Solicitation and Exploitation	Not Agree	Sexual Solicitation and Exploitation	Preadator is gradually escallating with sexual content and try to lower the victim's boundries to manipulate her into other motivations.
221	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator used flirtatious appearance-focused comments and private-channel invitations to build exclusivity, aiming to normalize inappropriate attention and leverage gifts for continued engagement.
222	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	None	None	Agree	None	This interaction shows no grooming behaviors; aside from one mild attempt to move communication to private messaging, the exchange reflects typical peer-level chatting and dares.
223	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator repeatedly sexualizes the child and requests explicit viewing to lower boundaries and gain access, aiming to escalate the interaction into increasingly exploitative behavior.
224	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator sexualizes the child and pushes for private visual access to lower boundaries and gain compliance, aiming to escalate the interaction into exploitative exchanges.
225	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator pushes escalating sexual acts and uses payment offers to secure compliance, aiming to gain full control over the child's sexualized behavior.

226	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator builds emotional closeness and shifts communication privately to strengthen exclusive bonds, aiming to increase influence and prepare for deeper boundary-testing later.
227	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator directs explicit sexual acts and pushes for private contact to intensify compliance, aiming to gain full access and escalate the exploitative interaction.
228	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator isolates the child and issues escalating sexual commands to gain exclusive control, aiming to create a private environment for continued exploitative behavior.
229	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Control and coercion	Control and coercion	Agree	Control and coercion	The preadator pushes aggressively for private communication and repeated contact to isolate the child, aiming to gain exclusive access and increase control over future interactions.
230	Control and Coersion	Control and Coersion	Agree	Control and Coersion	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator uses threats, pressure, and sexual suggestions to dominate the interaction, aiming to control the child's behavior and push toward private, escalating contact.
231	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator pressures the child for explicit exposure and frames it as a relationship obligation, aiming to secure compliance and escalate sexual access.
232	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	The preadator mixes sexual dares with pressure for private contact to lower boundaries and shift the interaction off-platform, aiming to escalate future sexual access.
233	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Control and coercion	Control and coercion	Agree	Control and coercion	The preadator pressures the child to reveal themselves and block others to gain influence, aiming to restrict their social space and increase personal control.
234	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	The preadator pressures the child for private contact and offers gifts to influence compliance, aiming to isolate them and escalate toward more exploitative interactions.
235	Control and coercion	Control and coercion	Agree	Control and coercion	None	None	Agree	None	The preadator uses threats, guilt, and escalating pressure to force compliance in the coin exchange, aiming to control the child's actions and maintain dominance.
236	None	None	Agree	None	None	None	Agree	None	There is no preadator behavior in this interaction; no grooming strategies are used, and no actions appear intended to influence, isolate, or sexualize the child.
237	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator issues escalating sexual commands and pushes for private contact to gain access and compliance, aiming to intensify control and continue exploitative behavior.
238	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	Agree	Sexual Solicitation and Exploitation	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator offers large rewards to pressure the child into sexual exposure, using escalating requests to exploit their desire for in-app status and compliance.
239	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Sexual Desensitization	Sexual Desensitization	Agree	Sexual Desensitization	The preadator introduces mild sexualized imagery and pushes for private communication, aiming to shift the interaction off-platform and test boundaries for potential escalation later.
240	Isolation and Secrecy	Isolation and Secrecy	Agree	Isolation and Secrecy	Control and coercion	Control and coercion	Agree	Control and coercion	The preadator uses affectionate language and pushes for private communication to isolate the child, aiming to build exclusive access and prepare for future boundary-testing.

Appendix D

Results

Rater Agreement

Cohen's kappa

			95% CI	
Ratings	kappa	SE	Lower	Upper
Average kappa	0.787			
Annotator 1 - Annotator 2	0.787	0.076	0.638	0.937

Note. 40 subjects/items and 2 raters/measurements. Based on pairwise complete cases. Confidence intervals are asymptotic.

Fleiss' kappa

			95% CI	
Ratings	Fleiss' kappa	SE	Lower	Upper
Overall	0.786	0.096	0.599	0.973
4	0.774	0.160	0.460	1.088
1	0.892	0.160	0.577	1.207
3	0.805	0.160	0.491	1.119
5	0.623	0.160	0.309	0.937
2	−0.013	0.160	−0.327	0.301

Note. 39 subjects/items and 2 raters/measurements. Based on listwise complete cases. Confidence intervals are asymptotic.

Krippendorff's alpha

			95% CI	
Method	Krippendorff's alpha	SE	Lower	Upper
Nominal	0.789	0.081	0.614	0.927

Note. 40 subjects/items and 2 raters/measurements. Based on pairwise complete cases.

Results

Rater Agreement

Cohen's kappa

			95% CI	
Ratings	kappa	SE	Lower	Upper
Average kappa	0.772			
Annotator_1 - Annotator_2	0.772	0.082	0.611	0.933

Note. 38 subjects/items and 2 raters/measurements. Based on pairwise complete cases. Confidence intervals are asymptotic.

Fleiss' kappa

			95% CI	
Ratings	Fleiss' kappa	SE	Lower	Upper
Overall	0.770	0.092	0.590	0.950
5	0.858	0.171	0.522	1.194
1	0.743	0.172	0.407	1.079
2	0.531	0.171	0.195	0.867
4	0.849	0.172	0.513	1.185
3	0.784	0.171	0.448	1.120

Note. 34 subjects/items and 2 raters/measurements. Based on listwise complete cases. Confidence intervals are asymptotic.

Krippendorff's alpha

			95% CI	
Method	Krippendorff's alpha	SE	Lower	Upper
Nominal	0.774	0.085	0.590	0.925

Note. 40 subjects/items and 2 raters/measurements. Based on pairwise complete cases.

Appendix E

```
CONDITION A — EVALUATION REPORT
Model: google/gemma-3-4b-it | Timestamp: 2026-05-01 11:25:54

Overall Accuracy: 0.4000  (40.0%)

Strategy                                      Prec    Rec     F1  Support
------------------------------------------------------------------------
Sexual Desensitization                      0.3750 0.2143 0.2727       14
Sexual Normalization                        0.0000 0.0000 0.0000        4
Isolation and Secrecy                       0.0000 0.0000 0.0000        1
Sexual Solicitation and Exploitation        0.7500 0.6316 0.6857       19
Control and Coercion                        0.0625 0.5000 0.1111        2
------------------------------------------------------------------------
MACRO AVERAGE                               0.2375 0.2692 0.2139       40

★  Macro-averaged F1: 0.2139

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     3     0     0     3     8
   SN     0     0     0     1     3
   IS     0     0     0     0     1
  SSE     4     0     0    12     3
   CC     1     0     0     0     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                    Gold   Pred
------------------------------------------------------------
Sexual Desensitization                        14      8
Sexual Normalization                           4      0
Isolation and Secrecy                          1      0
Sexual Solicitation and Exploitation          19     16
Control and Coercion                           2     16

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------------
-----------------------
conv_004.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_006.txt       Sexual Normalization                  Control and
Coercion                     ✗
conv_011.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_020.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_032.txt       Sexual Normalization                  Control and
Coercion                     ✗
conv_037.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_038.txt       Sexual Desensitization                Sexual
Desensitization              ✓
```

conv_058.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_067.txt	Control and Coercion	Control and Coercion	✓
conv_068.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_071.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_072.txt	Sexual Desensitization	Control and Coercion	✗
conv_075.txt	Sexual Desensitization	Control and Coercion	✗
conv_079.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_085.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_089.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_098.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_099.txt	Sexual Desensitization	Control and Coercion	✗
conv_101.txt	Sexual Desensitization	Control and Coercion	✗
conv_102.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_106.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_112.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_118.txt	Sexual Desensitization	Control and Coercion	✗
conv_133.txt	Sexual Desensitization	Control and Coercion	✗
conv_138.txt	Isolation and Secrecy	Control and Coercion	✗
conv_143.txt	Sexual Desensitization	Control and Coercion	✗
conv_146.txt	Sexual Normalization	Sexual Solicitation and Exploitation	✗
conv_149.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_150.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_160.txt	Sexual Desensitization	Control and Coercion	✗
conv_161.txt	Sexual Normalization	Control and Coercion	✗
conv_163.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_166.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_176.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_181.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓

```
conv_183.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_187.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_195.txt       Control and Coercion                   Sexual
Desensitization              ✗
conv_196.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓

── Full Classification Report ──────────────────────────────
                                  precision    recall  f1-score   support

              Sexual Desensitization     0.3750    0.2143    0.2727        14
                Sexual Normalization     0.0000    0.0000    0.0000         4
               Isolation and Secrecy     0.0000    0.0000    0.0000         1
Sexual Solicitation and Exploitation     0.7500    0.6316    0.6857        19
                Control and Coercion     0.0625    0.5000    0.1111         2

                            accuracy                         0.4000        40
                           macro avg     0.2375    0.2692    0.2139        40
                        weighted avg     0.4906    0.4000    0.4267        40
```

Appendix F

```
CONDITION B — EVALUATION REPORT
Model: google/gemma-3-4b-it | Timestamp: 2026-05-04 08:23:13

Overall Accuracy: 0.5250  (52.5%)

Strategy                                      Prec    Rec     F1  Support
------------------------------------------------------------------------
Sexual Desensitization                      0.5385 0.5000 0.5185       14
Sexual Normalization                        1.0000 0.2500 0.4000        4
Isolation and Secrecy                       0.0000 0.0000 0.0000        1
Sexual Solicitation and Exploitation        0.5652 0.6842 0.6190       19
Control and Coercion                        0.0000 0.0000 0.0000        2
------------------------------------------------------------------------
MACRO AVERAGE                               0.3506 0.2390 0.2563       40

★  Macro-averaged F1: 0.2563

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     7     0     0     7     0
   SN     2     1     0     1     0
   IS     0     0     0     0     1
  SSE     4     0     0    13     1
   CC     0     0     0     2     0
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                    Gold   Pred
-------------------------------------------------------
Sexual Desensitization                        14     13
Sexual Normalization                           4      1
Isolation and Secrecy                          1      0
Sexual Solicitation and Exploitation          19     23
Control and Coercion                           2      2

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
-------------------------------------------------------------------------------
-----------------------
conv_004.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_006.txt       Sexual Normalization                  Sexual Solicitation
and Exploitation   ✗
conv_011.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_020.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_032.txt       Sexual Normalization                  Sexual
Desensitization              ✗
conv_037.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_038.txt       Sexual Desensitization                Sexual
Desensitization              ✓
```

conv_058.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_067.txt	Control and Coercion	Sexual Solicitation and Exploitation	✗
conv_068.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_071.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_072.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_075.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_079.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_085.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_089.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_098.txt	Sexual Solicitation and Exploitation	control_coercion	✗
conv_099.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_101.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_102.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_106.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_112.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_118.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_133.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_138.txt	Isolation and Secrecy	Control and Coercion	✗
conv_143.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_146.txt	Sexual Normalization	Sexual Normalization	✓
conv_149.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_150.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_160.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_161.txt	Sexual Normalization	Sexual Desensitization	✗
conv_163.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_166.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_176.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_181.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓

```
conv_183.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_187.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_195.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗
conv_196.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓

── Full Classification Report ──────────────────────────────
                                   precision    recall  f1-score   support

              Sexual Desensitization     0.5385    0.5000    0.5185        14
                Sexual Normalization     1.0000    0.2500    0.4000         4
               Isolation and Secrecy     0.0000    0.0000    0.0000         1
Sexual Solicitation and Exploitation     0.5652    0.6842    0.6190        19
                 Control and Coercion     0.0000    0.0000    0.0000         2

                           micro avg     0.5385    0.5250    0.5316        40
                           macro avg     0.4207    0.2868    0.3075        40
                        weighted avg     0.5569    0.5250    0.5155        40
```

Appendix G

```
CONDITION A — EVALUATION REPORT
Model: google/gemma-3-4b-it | Timestamp: 2026-05-01 11:25:54

Overall Accuracy: 0.4000  (40.0%)

Strategy                                      Prec    Rec     F1  Support
------------------------------------------------------------------------
Sexual Desensitization                      0.3750 0.2143 0.2727       14
Sexual Normalization                        0.0000 0.0000 0.0000        4
Isolation and Secrecy                       0.0000 0.0000 0.0000        1
Sexual Solicitation and Exploitation        0.7500 0.6316 0.6857       19
Control and Coercion                        0.0625 0.5000 0.1111        2
------------------------------------------------------------------------
MACRO AVERAGE                               0.2375 0.2692 0.2139       40

★  Macro-averaged F1: 0.2139

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     3     0     0     3     8
   SN     0     0     0     1     3
   IS     0     0     0     0     1
  SSE     4     0     0    12     3
   CC     1     0     0     0     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                    Gold   Pred
------------------------------------------------------------
Sexual Desensitization                        14      8
Sexual Normalization                           4      0
Isolation and Secrecy                          1      0
Sexual Solicitation and Exploitation          19     16
Control and Coercion                           2     16

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------------
-----------------------
conv_004.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_006.txt       Sexual Normalization                  Control and
Coercion                     ✗
conv_011.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_020.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_032.txt       Sexual Normalization                  Control and
Coercion                     ✗
conv_037.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_038.txt       Sexual Desensitization                Sexual
Desensitization              ✓
```

conv_058.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_067.txt	Control and Coercion	Control and Coercion	✓
conv_068.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_071.txt	Sexual Desensitization	Sexual Desensitization	✓
conv_072.txt	Sexual Desensitization	Control and Coercion	✗
conv_075.txt	Sexual Desensitization	Control and Coercion	✗
conv_079.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_085.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_089.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_098.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_099.txt	Sexual Desensitization	Control and Coercion	✗
conv_101.txt	Sexual Desensitization	Control and Coercion	✗
conv_102.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_106.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_112.txt	Sexual Desensitization	Sexual Solicitation and Exploitation	✗
conv_118.txt	Sexual Desensitization	Control and Coercion	✗
conv_133.txt	Sexual Desensitization	Control and Coercion	✗
conv_138.txt	Isolation and Secrecy	Control and Coercion	✗
conv_143.txt	Sexual Desensitization	Control and Coercion	✗
conv_146.txt	Sexual Normalization	Sexual Solicitation and Exploitation	✗
conv_149.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_150.txt	Sexual Solicitation and Exploitation	Sexual Desensitization	✗
conv_160.txt	Sexual Desensitization	Control and Coercion	✗
conv_161.txt	Sexual Normalization	Control and Coercion	✗
conv_163.txt	Sexual Solicitation and Exploitation	Control and Coercion	✗
conv_166.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_176.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓
conv_181.txt	Sexual Solicitation and Exploitation	Sexual Solicitation and Exploitation	✓

```
conv_183.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_187.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_195.txt       Control and Coercion                   Sexual
Desensitization              ✗
conv_196.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓

── Full Classification Report ──────────────────────────────
                                  precision    recall  f1-score   support

              Sexual Desensitization     0.3750    0.2143    0.2727        14
                Sexual Normalization     0.0000    0.0000    0.0000         4
              Isolation and Secrecy      0.0000    0.0000    0.0000         1
Sexual Solicitation and Exploitation     0.7500    0.6316    0.6857        19
                Control and Coercion     0.0625    0.5000    0.1111         2

                            accuracy                         0.4000        40
                           macro avg     0.2375    0.2692    0.2139        40
                        weighted avg     0.4906    0.4000    0.4267        40
```

```
CONDITION B — EVALUATION REPORT
Model: google/gemma-3-4b-it | Timestamp: 2026-04-04 10:44:10

Overall Accuracy: 0.4750  (47.5%)

Strategy                                        Prec    Rec     F1  Support
```

```
------------------------------------------------------------------------
Sexual Desensitization                       0.5000 0.2857 0.3636       14
Sexual Normalization                         0.5000 0.2500 0.3333        4
Isolation and Secrecy                        0.0000 0.0000 0.0000        1
Sexual Solicitation and Exploitation         0.5417 0.6842 0.6047       19
Control and Coercion                         0.1667 0.5000 0.2500        2
------------------------------------------------------------------------
MACRO AVERAGE                                0.3417 0.3440 0.3103       40

★  Macro-averaged F1: 0.3103

── Confusion Matrix (rows=gold, cols=predicted) ─────────────
         SD    SN    IS   SSE    CC
   SD     4     1     0     8     1
   SN     0     1     0     2     1
   IS     0     0     0     0     1
  SSE     4     0     0    13     2
   CC     0     0     0     1     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ──────────────────────────────────────
Strategy                                          Gold   Pred
------------------------------------------------------------
Sexual Desensitization                              14      8
Sexual Normalization                                 4      2
Isolation and Secrecy                                1      0
Sexual Solicitation and Exploitation                19     24
Control and Coercion                                 2      6

── Per-Conversation Results ────────────────────────────────
Filename           Gold Primary                           Pred Primary
Match
------------------------------------------------------------------------
----------------------
conv_004.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_006.txt       Sexual Normalization                   Sexual Solicitation
and Exploitation   ✗
conv_011.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_020.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_032.txt       Sexual Normalization                   Control and
Coercion                   ✗
conv_037.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_038.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_058.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_067.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗
conv_068.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
```

```
conv_071.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_072.txt       Sexual Desensitization               Sexual
Normalization              ✗
conv_075.txt       Sexual Desensitization               Sexual
Desensitization            ✓
conv_079.txt       Sexual Solicitation and Exploitation Sexual
Desensitization            ✗
conv_085.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_089.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_098.txt       Sexual Solicitation and Exploitation Control and
Coercion                   ✗
conv_099.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_101.txt       Sexual Desensitization               Sexual
Desensitization            ✓
conv_102.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_106.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_112.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_118.txt       Sexual Desensitization               Sexual
Desensitization            ✓
conv_133.txt       Sexual Desensitization               Sexual
Desensitization            ✓
conv_138.txt       Isolation and Secrecy                Control and
Coercion                   ✗
conv_143.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_146.txt       Sexual Normalization                 Sexual Solicitation
and Exploitation   ✗
conv_149.txt       Sexual Solicitation and Exploitation Sexual
Desensitization            ✗
conv_150.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_160.txt       Sexual Desensitization               Control and
Coercion                   ✗
conv_161.txt       Sexual Normalization                 Sexual
Normalization              ✓
conv_163.txt       Sexual Solicitation and Exploitation Control and
Coercion                   ✗
conv_166.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_176.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_181.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_183.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation Sexual Solicitation
and Exploitation   ✓
conv_187.txt       Sexual Solicitation and Exploitation Sexual
Desensitization            ✗
```

```
conv_195.txt       Control and Coercion                  Control and
Coercion                  ✓
conv_196.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓

── Full Classification Report ──────────────────────────────
                                  precision    recall  f1-score   support

             Sexual Desensitization     0.5000    0.2857    0.3636        14
               Sexual Normalization     0.5000    0.2500    0.3333         4
              Isolation and Secrecy     0.0000    0.0000    0.0000         1
Sexual Solicitation and Exploitation     0.5417    0.6842    0.6047        19
                Control and Coercion     0.1667    0.5000    0.2500         2

                            accuracy                         0.4750        40
                           macro avg     0.3417    0.3440    0.3103        40
                        weighted avg     0.4906    0.4750    0.4603        40
```

Appendix H

```
CONDITION A — EVALUATION REPORT
Model: google/gemma-3-12b-it | Timestamp: 2026-05-08 18:34:41

Overall Accuracy: 0.6250  (62.5%)

Strategy                                        Prec    Rec     F1  Support
------------------------------------------------------------------------------
Sexual Desensitization                        1.0000 0.1429 0.2500        7
Sexual Normalization                          1.0000 0.3333 0.5000        3
Isolation and Secrecy                         0.5000 1.0000 0.6667        1
Sexual Solicitation and Exploitation          0.6562 0.8400 0.7368       25
Control and Coercion                          0.3333 0.2500 0.2857        4
------------------------------------------------------------------------------
MACRO AVERAGE                                 0.5816 0.4277 0.4065       40

★  Macro-averaged F1: 0.4065

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     1     0     0     6     0
   SN     0     1     0     2     0
   IS     0     0     1     0     0
  SSE     0     0     1    21     2
   CC     0     0     0     3     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                       Gold   Pred
-------------------------------------------------------------
Sexual Desensitization                            7      1
Sexual Normalization                              3      1
Isolation and Secrecy                             1      2
Sexual Solicitation and Exploitation             25     32
Control and Coercion                              4      3

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------------
-----------------------
conv_001.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_014.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_019.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_025.txt       Sexual Solicitation and Exploitation   Isolation and
Secrecy                  ✗
conv_033.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_035.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_047.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
```

```
conv_051.txt       Sexual Normalization                 Sexual Solicitation
and Exploitation   ✗
conv_052.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_056.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_061.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_074.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_075.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_077.txt       Sexual Solicitation and Exploitation   Control and
Coercion                ✗
conv_087.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_100.txt       Control and Coercion                 Control and
Coercion                ✓
conv_103.txt       Sexual Normalization                 Sexual
Normalization                ✓
conv_111.txt       Control and Coercion                 Sexual Solicitation
and Exploitation   ✗
conv_115.txt       Sexual Desensitization               Sexual
Desensitization              ✓
conv_116.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_120.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_121.txt       Sexual Normalization                 Sexual Solicitation
and Exploitation   ✗
conv_130.txt       Isolation and Secrecy                Isolation and
Secrecy                 ✓
conv_132.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_135.txt       Sexual Solicitation and Exploitation   Control and
Coercion                ✗
conv_142.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_145.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_151.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_157.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_159.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_164.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_169.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_173.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_178.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_179.txt       Sexual Solicitation and Exploitation   None
✗
```

```
conv_180.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_189.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗
conv_194.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_195.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗

── Full Classification Report ──────────────────────────────
                                   precision    recall  f1-score   support

              Sexual Desensitization     1.0000    0.1429    0.2500         7
                Sexual Normalization     1.0000    0.3333    0.5000         3
              Isolation and Secrecy      0.5000    1.0000    0.6667         1
Sexual Solicitation and Exploitation     0.6562    0.8400    0.7368        25
                Control and Coercion     0.3333    0.2500    0.2857         4

                         micro avg     0.6410    0.6250    0.6329        40
                         macro avg     0.6979    0.5132    0.4878        40
                      weighted avg     0.7060    0.6250    0.5870        40
```

```
CONDITION B — EVALUATION REPORT
Model: google/gemma-3-12b-it | Timestamp: 2026-05-08 18:56:11

Overall Accuracy: 0.6000  (60.0%)

Strategy                                    Prec    Rec     F1  Support
------------------------------------------------------------------------
Sexual Desensitization                    0.5000 0.1429 0.2222        7
```

```
Sexual Normalization                          1.0000 0.3333 0.5000        3
Isolation and Secrecy                         0.5000 1.0000 0.6667        1
Sexual Solicitation and Exploitation          0.6452 0.8000 0.7143       25
Control and Coercion                          0.3333 0.2500 0.2857        4
------------------------------------------------------------------------------
MACRO AVERAGE                                 0.4964 0.4210 0.3981       40

★  Macro-averaged F1: 0.3981

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     1     0     0     6     0
   SN     0     1     0     2     0
   IS     0     0     1     0     0
  SSE     1     0     1    20     2
   CC     0     0     0     3     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                        Gold   Pred
------------------------------------------------------------
Sexual Desensitization                             7      2
Sexual Normalization                               3      1
Isolation and Secrecy                              1      2
Sexual Solicitation and Exploitation              25     31
Control and Coercion                               4      3

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------------
-----------------------
conv_001.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_014.txt       Sexual Solicitation and Exploitation   Sexual
Desensitization              ✗
conv_019.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_025.txt       Sexual Solicitation and Exploitation   Isolation and
Secrecy                    ✗
conv_033.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_035.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_047.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_051.txt       Sexual Normalization                   Sexual Solicitation
and Exploitation   ✗
conv_052.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_056.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_061.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
```

```
conv_074.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_075.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_077.txt       Sexual Solicitation and Exploitation   Control and
Coercion                  ✗
conv_087.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_100.txt       Control and Coercion                   Control and
Coercion                  ✓
conv_103.txt       Sexual Normalization                   Sexual
Normalization                ✓
conv_111.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗
conv_115.txt       Sexual Desensitization                 Sexual
Desensitization              ✓
conv_116.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_120.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_121.txt       Sexual Normalization                   Sexual Solicitation
and Exploitation   ✗
conv_130.txt       Isolation and Secrecy                  Isolation and
Secrecy                   ✓
conv_132.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_135.txt       Sexual Solicitation and Exploitation   Control and
Coercion                  ✗
conv_142.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_145.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_151.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_157.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_159.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_164.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_169.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_173.txt       Sexual Desensitization                 Sexual Solicitation
and Exploitation   ✗
conv_178.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_179.txt       Sexual Solicitation and Exploitation   None
✗
conv_180.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_184.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_189.txt       Control and Coercion                   Sexual Solicitation
and Exploitation   ✗
conv_194.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
```

```
conv_195.txt       Control and Coercion                  Sexual Solicitation
and Exploitation    ✕

── Full Classification Report ──────────────────────────────
                                   precision    recall  f1-score   support

              Sexual Desensitization     0.5000    0.1429    0.2222         7
                Sexual Normalization     1.0000    0.3333    0.5000         3
               Isolation and Secrecy     0.5000    1.0000    0.6667         1
Sexual Solicitation and Exploitation     0.6452    0.8000    0.7143        25
                Control and Coercion     0.3333    0.2500    0.2857         4

                           micro avg     0.6154    0.6000    0.6076        40
                           macro avg     0.5957    0.5052    0.4778        40
                        weighted avg     0.6116    0.6000    0.5681        40
```

Appendix I

```
Gemma3-4B
CONDITION B — Cross-Platform EVALUATION REPORT
Model: google/gemma-3-4b-it | Timestamp: 2026-05-08 20:17:38
Gold source: data/Fine_tune_data_sheet.xlsx

Overall Accuracy: 0.4359  (43.6%)

Strategy                                        Prec    Rec     F1  Support
------------------------------------------------------------------------
Sexual Desensitization                        0.6667 0.2857 0.4000       14
Sexual Normalization                          0.0000 0.0000 0.0000        0
Isolation and Secrecy                         1.0000 0.3636 0.5333       11
Sexual Solicitation and Exploitation          0.3636 0.8000 0.5000       10
Control and Coercion                          0.2500 0.2500 0.2500        4
------------------------------------------------------------------------
MACRO AVERAGE                                 0.4561 0.3399 0.3367       39

★  Macro-averaged F1: 0.3367

── Confusion Matrix (rows=gold, cols=predicted) ──────────
         SD    SN    IS   SSE    CC
   SD     4     0     0     8     1
   SN     0     0     0     0     0
   IS     1     0     4     4     1
  SSE     1     0     0     8     1
   CC     0     0     0     2     1
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ────────────────────────────────────
Strategy                                      Gold   Pred
------------------------------------------------------------
Sexual Desensitization                          14      6
Sexual Normalization                             0      0
Isolation and Secrecy                           11      4
Sexual Solicitation and Exploitation            10     22
Control and Coercion                             4      4

── Per-Conversation Results ──────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------
----------------------
conv_201.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_202.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_203.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_204.txt       Sexual Desensitization                Sexual
Desensitization                ✓
conv_205.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_206.txt       Sexual Desensitization                Sexual
Desensitization                ✓
```

```
conv_207.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_208.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_209.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_210.txt       Sexual Solicitation and Exploitation  Control and
Coercion                ✗
conv_211.txt       Sexual Desensitization                None
✗
conv_212.txt       Sexual Solicitation and Exploitation  Sexual
Desensitization              ✗
conv_213.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_214.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_215.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_216.txt       Isolation and Secrecy                 Isolation and
Secrecy                 ✓
conv_217.txt       Control and Coercion                  None
✗
conv_218.txt       Control and Coercion                  Control and
Coercion                ✓
conv_219.txt       Isolation and Secrecy                 Sexual Solicitation
and Exploitation   ✗
conv_220.txt       Isolation and Secrecy                 Sexual Solicitation
and Exploitation   ✗
conv_221.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_222.txt       Isolation and Secrecy                 Isolation and
Secrecy                 ✓
conv_223.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_224.txt       Sexual Desensitization                Control and
Coercion                ✗
conv_225.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_226.txt       Isolation and Secrecy                 Sexual
Desensitization              ✗
conv_227.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_228.txt       Isolation and Secrecy                 Control and
Coercion                ✗
conv_229.txt       Isolation and Secrecy                 None
✗
conv_230.txt       Control and Coercion                  Sexual Solicitation
and Exploitation   ✗
conv_231.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_232.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_233.txt       Isolation and Secrecy                 Isolation and
Secrecy                 ✓
conv_234.txt       Isolation and Secrecy                 Sexual Solicitation
and Exploitation   ✗
```

```
conv_235.txt       Control and Coercion                  Sexual Solicitation
and Exploitation   ✗
conv_236.txt       None                                  Sexual
Desensitization              ✗
conv_237.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_238.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_239.txt       Isolation and Secrecy                 Sexual Solicitation
and Exploitation   ✗
conv_240.txt       Isolation and Secrecy                 Isolation and
Secrecy                  ✓

── Full Classification Report ──────────────────────────────
                                   precision    recall  f1-score   support

              Sexual Desensitization     0.6667    0.2857    0.4000        14
                Sexual Normalization     0.0000    0.0000    0.0000         0
               Isolation and Secrecy     1.0000    0.3636    0.5333        11
Sexual Solicitation and Exploitation     0.3636    0.8000    0.5000        10
                Control and Coercion     0.2500    0.2500    0.2500         4

                           micro avg     0.4722    0.4359    0.4533        39
                           macro avg     0.4561    0.3399    0.3367        39
                        weighted avg     0.6402    0.4359    0.4479        39
```

Gemma 3-12B

```
CONDITION B —Cross Platform EVALUATION REPORT
Model: google/gemma-3-12b-it | Timestamp: 2026-05-08 19:38:39
Gold source: data/Fine_tune_data_sheet.xlsx
```

```
Overall Accuracy: 0.5385  (53.8%)

Strategy                                        Prec    Rec     F1  Support
------------------------------------------------------------------------------
Sexual Desensitization                        1.0000 0.2857 0.4444       14
Sexual Normalization                          0.0000 0.0000 0.0000        0
Isolation and Secrecy                         0.8333 0.4545 0.5882       11
Sexual Solicitation and Exploitation          0.4444 0.8000 0.5714       10
Control and Coercion                          0.4000 1.0000 0.5714        4
------------------------------------------------------------------------------
MACRO AVERAGE                                 0.5356 0.5081 0.4351       39

★  Macro-averaged F1: 0.4351

── Confusion Matrix (rows=gold, cols=predicted) ─────────────
         SD    SN    IS   SSE    CC
   SD     4     0     1     7     1
   SN     0     0     0     0     0
   IS     0     0     5     3     3
  SSE     0     0     0     8     2
   CC     0     0     0     0     4
SD=Sexual Desensitization  SN=Sexual Normalization
IS=Isolation and Secrecy   SSE=Sexual Solicitation and Exploitation
CC=Control and Coercion

── Label Distribution ──────────────────────────────────────
Strategy                                      Gold   Pred
------------------------------------------------------------
Sexual Desensitization                          14      4
Sexual Normalization                             0      0
Isolation and Secrecy                           11      6
Sexual Solicitation and Exploitation            10     18
Control and Coercion                             4     10

── Per-Conversation Results ────────────────────────────────
Filename           Gold Primary                          Pred Primary
Match
------------------------------------------------------------------------------
-----------------------
conv_201.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_202.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_203.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
conv_204.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_205.txt       Sexual Desensitization                Control and
Coercion                     ✗
conv_206.txt       Sexual Desensitization                Sexual
Desensitization              ✓
conv_207.txt       Sexual Desensitization                Sexual Solicitation
and Exploitation   ✗
conv_208.txt       Sexual Solicitation and Exploitation  Sexual Solicitation
and Exploitation   ✓
```

```
conv_209.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_210.txt       Sexual Solicitation and Exploitation   Control and
Coercion                  ✗
conv_211.txt       Sexual Desensitization               None
✗
conv_212.txt       Sexual Solicitation and Exploitation   Control and
Coercion                  ✗
conv_213.txt       Sexual Desensitization               Sexual
Desensitization                ✓
conv_214.txt       Sexual Desensitization               Isolation and
Secrecy                   ✗
conv_215.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_216.txt       Isolation and Secrecy                Isolation and
Secrecy                   ✓
conv_217.txt       Control and Coercion                 Control and
Coercion                  ✓
conv_218.txt       Control and Coercion                 Control and
Coercion                  ✓
conv_219.txt       Isolation and Secrecy                Sexual Solicitation
and Exploitation   ✗
conv_220.txt       Isolation and Secrecy                Sexual Solicitation
and Exploitation   ✗
conv_221.txt       Sexual Desensitization               Sexual
Desensitization                ✓
conv_222.txt       Isolation and Secrecy                Control and
Coercion                  ✗
conv_223.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_224.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_225.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_226.txt       Isolation and Secrecy                Control and
Coercion                  ✗
conv_227.txt       Sexual Desensitization               Sexual Solicitation
and Exploitation   ✗
conv_228.txt       Isolation and Secrecy                Isolation and
Secrecy                   ✓
conv_229.txt       Isolation and Secrecy                Isolation and
Secrecy                   ✓
conv_230.txt       Control and Coercion                 Control and
Coercion                  ✓
conv_231.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_232.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_233.txt       Isolation and Secrecy                Isolation and
Secrecy                   ✓
conv_234.txt       Isolation and Secrecy                Sexual Solicitation
and Exploitation   ✗
conv_235.txt       Control and Coercion                 Control and
Coercion                  ✓
conv_236.txt       None                                 None
✓
```

```
conv_237.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_238.txt       Sexual Solicitation and Exploitation   Sexual Solicitation
and Exploitation   ✓
conv_239.txt       Isolation and Secrecy                  Control and
Coercion                  ✗
conv_240.txt       Isolation and Secrecy                  Isolation and
Secrecy                   ✓

── Full Classification Report ──────────────────────────────
                                  precision    recall  f1-score   support

              Sexual Desensitization     1.0000    0.2857    0.4444        14
                Sexual Normalization     0.0000    0.0000    0.0000         0
               Isolation and Secrecy     0.8333    0.4545    0.5882        11
Sexual Solicitation and Exploitation     0.4444    0.8000    0.5714        10
                Control and Coercion     0.4000    1.0000    0.5714         4

                           micro avg     0.5526    0.5385    0.5455        39
                           macro avg     0.5356    0.5081    0.4351        39
                        weighted avg     0.7490    0.5385    0.5306        39
```